

Digital (Killer?) Acquisitions*

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Abstract

This paper examines innovation outcomes before and after 1,200 startup acquisitions by eight major technology firms. Linking patent and workforce data to these deals, we document four main findings. First, most acquired startups hold no patents, but those with patents tend to operate in technology areas where the acquirer already has a presence and that see further acquisition activity. Second, innovation typically rises before an acquisition, continues afterward only where further acquisitions follow, and falls back once acquisition activity ends. We propose a stylized model in which beliefs about commercial viability drive both startup entry and acquisitions to explain how these patterns arise through selection rather than effects of the deals. Third, acquired patents receive significantly more citations after the acquisition than comparable patents, not only from the acquirer but also from firms that hire the targets' employees. Fourth, although 31% of employees and 23% of inventors depart within a year, inventors who stay go on to patent substantially more, while stayers and leavers were equally productive beforehand. In the aggregate, we find little evidence that acquisitions by digital incumbents suppress innovation, even though serial acquisitions go hand in hand with growing patent consolidation.

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1 Introduction

The rising dominance of large digital platforms has sparked renewed interest in the competitive implications of their acquisition behavior. Over the past two decades, leading technology firms—Amazon, Apple, Facebook, Google, and Microsoft (commonly referred to as GAFAM), as well as Cisco, Intel, and Qualcomm—have each engaged in hundreds of acquisitions, often targeting early-stage startups in adjacent or emerging technology markets. Many of these purchases have come in series with the same firm buying several startups in the same technology area within a few years.¹ These transactions raise important questions about how markets for technology function, and whether acquisitions help or hinder the diffusion and commercialization of inventions.

Given the complementary assets required to bring new technologies to market, innovative startups often rely on licensing deals or acquisitions to appropriate returns from their inventions (Teece, 1986; Gans and Stern, 2000; Gans, Hsu and Stern, 2002; Arora, Fosfuri and Gambardella, 2001). At the same time, regulators, policymakers, and scholars have raised concerns that some of these acquisitions may serve anticompetitive purposes. A growing body of research emphasizes innovation-based theories of harm, highlighting that mergers can reduce innovation incentives, eliminate nascent competition, or alter the direction of technological progress (Federico, Langus and Valletti, 2017, 2018; Federico, Scott Morton and Shapiro, 2020; Cunningham, Ederer and Ma, 2021; Valletti, 2025). These concerns are particularly salient in digital markets, where acquisitions can affect not only current competition but also the trajectory of future innovation. Recent policy discussions and interventions, including the 2023 U.S. Merger Guidelines and the EU Digital Markets Act, explicitly recognize that repeated or serial acquisitions may cumulatively undermine competitive innovation dynamics.²

This paper investigates the path of innovation within narrowly defined technological domains before and after a series of startup acquisitions by major digital incumbents. Instead of focusing on the ex-ante incentives of potential acquisition targets which is a common

¹For example, Cisco famously built its networking business through a steady stream of acquisitions in the 1990s (Cisco Systems, 2007). Google entered mapping and later artificial intelligence through successions of startup purchases (Google, 2004; Oreskovic and Roumeliotis, 2014). Former Apple CEO Tim Cook stated that the company acquires a company, on average, every two to three weeks (Feiner, 2019).

²Concerns about acquisitions that may lead to the termination of overlapping innovation projects were first raised in the pharmaceutical industry (Cunningham et al., 2021), but have since been applied to the digital economy (OECD, 2020; Federal Trade Commission, 2021) and other R&D-based industries. However, in contrast to pharmaceuticals the innovation environment in the tech industry is less reliant on formal intellectual property and more centered around talent acquisition and platform integration (Gawer and Cusumano, 2002; Jin, Leccese and Wagman, 2023; Olmstead-Rumsey, Puglisi and Wu, 2024).

emphasis in the theoretical literature,³ we examine ex-post outcomes using a series of event studies. Specifically, we analyze changes in patenting and patent citations before and after an acquisition, thereby assessing how these deals reshape the trajectory of innovation in the affected technology space.

We make three main contributions to the literature on acquisitions, innovation, and competition policy. First, we provide a comprehensive empirical mapping of the acquisition behavior of eight major digital firms (the GAFAM firms plus Cisco, Intel and Qualcomm). We manually compile a dataset of roughly 1,200 acquisitions and link this information to patent data from PatentsView and labor market data from Revelio Labs. We show that while these firms have acquired a large number of startups, the majority of targets do not hold patents at the time of acquisition which suggests that many deals are motivated by access to capabilities, products, customers, or personnel rather than formal intellectual property. We also find that while retention rates vary substantially across deals, on average 31% of target employees and 23% of their inventors leave the acquirer within a year of the acquisition, highlighting the fragility of workforce integration.

We find that acquisitions are highly concentrated in specific technological domains and often represent repeated engagement by the same acquirer in a given area. For instance, firms like Intel and Qualcomm which are traditionally hardware-focused, tend to acquire targets in Cooperative Patent Classification (CPC) categories where they already hold patents, suggesting a form of cumulative innovation or vertical integration. In contrast, software-oriented firms such as Facebook or Amazon are more likely to pursue acquisitions without patents, perhaps reflecting a focus on user bases, engineering talent, or data assets. As acquisitions accumulate in an area, the acquirers' combined share of its patents rises, so serial dealmaking goes hand in hand with a gradual consolidation of patent ownership.

Second, we use patent data to construct a panel at the technology-class-year level that allows us to analyze innovation trends surrounding each deal. We find that innovation frequently precedes acquisition: many acquisitions follow a surge of patenting within the same CPC groups where the acquired startup obtains patents, consistent with incumbents acquiring startups to access promising technologies. Using a suite of event-study analyses, we also study how innovation activity evolves after acquisitions. We find that patenting activity in affected CPC groups generally continues to rise after an acquisition. This pattern is robust to matching each CPC group with an acquisition to a comparison group on pre-acquisition growth trends and controlling for a range of fixed effects. However, the post-acquisition

³See, in addition to the previously mentioned contributions, Kamepalli, Rajan and Zingales (2020), Fumagalli, Motta and Tarantino (2020), Bryan and Hovenkamp (2020), Motta and Shelegia (2025), and Letina, Schmutzler and Seibel (2024).

growth in patenting is only present when follow-on acquisitions occur in the same technology space, and it partly unwinds once acquisition activity in an area ends.

To organize these patterns, we develop a stylized model of acquisitions in a technology space whose commercial viability is uncertain and gradually revealed through public signals. Because both an entrant’s incentive to develop a project and an incumbent’s incentive to acquire it move together with beliefs about viability, the model predicts that acquisitions cluster into waves, that innovation rises ahead of an acquisition in spaces that are subsequently acquired (through selection across technology states rather than through any causal effect of the deal), and that the incumbent’s accumulated presence keeps innovation persistently below the path it would have followed without the acquisitions. The model also delineates the conditions under which deals take the form of genuine killer acquisitions, in which the incumbent acquires a project only to shelve it, an outcome we cannot observe directly in the data. The pre-acquisition rise alone does not rule out killer motives, since a rising technology is exactly what a threatened incumbent would want to shut down, and the model produces killer acquisitions during belief-driven booms. At the same time, however, the empirical evidence fits poorly with a narrative in which incumbents frequently suppress technologies before they gain momentum.

Third, in addition to the CPC group-level analyses, we examine the effects of acquisitions at the level of individual patents and inventors. Using a difference-in-differences framework, we show that acquired patents receive significantly more citations after the acquisition than comparable patents in the same CPC group. These effects are not limited to citations from the acquiring firm, but extend to third-party citations, suggesting that acquisitions increase the visibility and influence of acquired technologies across the broader technology ecosystem. The post-acquisition citation boost is especially pronounced for first-acquired targets in CPC groups that experience follow-on acquisitions, indicating that the strategic sequencing of acquisitions may amplify knowledge diffusion. We also use the Revelio workforce data to decompose where these additional citations originate. We find no detectable relationship between the size of the post-acquisition citation increase and how many of the target’s inventors are retained at the acquirer, but its composition shifts. When retention is high, the acquirer itself accounts for more of the additional citations. But when retention is low, more come from firms that hire inventors away from the acquirer’s targets. This is what one would expect if the acquired knowledge traveled with the inventors, though retention is not randomly assigned and thus this pattern has to be interpreted with some caution. At the inventor level, target inventors who stay with the acquirer go on to patent substantially more than before the acquisition, while those who stayed and those who left were equally productive beforehand.

Our findings complicate the popular narrative that acquisitions by large digital incumbents predominantly suppress innovation. While we cannot rule out killer acquisitions in individual cases, since shelving is not observable in patent data, the outcomes we document are more consistent with complementarity than with suppression. Innovation tends to rise rather than fall in the wake of acquisition, though only in technological domains where the acquirer continues to make follow-on investments, which tend to be domains where it already held patents. Acquired patents are cited more, not less, after the deal, including by firms well beyond the acquirer, and target inventors who stay go on to patent more than before.

This paper contributes to several strands of literature at the intersection of innovation, acquisitions, and competition policy. First, we build on a growing body of empirical work on killer acquisitions, beginning with [Cunningham et al. \(2021\)](#) in the pharmaceutical industry. Their framework identifies acquisitions that lead to the termination of overlapping drug development projects, thereby suppressing future competition. Subsequent research has extended this idea to other sectors. For example, [Lemos and Resende \(2025\)](#) examine digital markets, while [Kamepalli et al. \(2020\)](#) model a “kill zone” in which the prospect of acquisition by a dominant platform lowers user adoption of entrants and venture funding for startups in the affected space. Our analysis complements this work by focusing on ex-post outcomes in patenting activity rather than ex-ante project terminations. [Jin et al. \(2023\)](#) show that GAFAM made more tech acquisitions per firm than other top acquirers, targeting younger and more consumer-facing firms. Crucially, they find no evidence that GAFAM acquisitions reduce entry by other acquirers in the same category, suggesting that such deals do not deter competition, thereby complementing our finding that innovation does not, on average, decline within targeted technologies after acquisition. [Gugler, Szücs and Wohak \(2025\)](#) also examine patenting in technology classes affected by GAFAM acquisitions, alongside venture capital investment in related markets. We show that acquisitions select into technology areas where innovation is already rising and examine acquisition waves and knowledge diffusion beyond the acquired firms.

Second, the paper connects to a broader literature on the link between competition policy and innovation ([Gilbert, 2020](#)). Our event-study design allows us to analyze such dynamics within narrowly defined technological areas. In contrast to most existing work, we focus specifically on startup acquisitions by digital incumbents and analyze repeated acquisitions over time. Like us, [Watzinger, Fackler, Nagler and Schnitzer \(2020\)](#) and [Poege \(2024\)](#) use patent data to study how competition policy influences innovation across a set of related technology domains. Whereas they focus on two historically significant enforcement actions (the 1956 AT&T consent decree and the post-war breakup of IG Farben, respectively), we consider a large number of more recent acquisitions.

Recent antitrust policy has also shifted toward recognizing the broader strategic implications of repeated acquisitions. Rather than evaluating transactions solely on a case-by-case basis, regulators are increasingly concerned with the cumulative effects of serial acquisitions within a particular technological or geographic domain (Jin, Leccese, Wagman and Wang, 2025). The 2023 U.S. Merger Guidelines explicitly acknowledge that mergers occurring as part of a pattern or roll-up strategy may be assessed collectively rather than in isolation (U.S. Department of Justice and Federal Trade Commission, 2023). This perspective is reinforced by recent empirical work on acquisition rollups in health care markets, which documents the anticompetitive risks posed by fragmented yet systematic buying strategies (Asil, Ramos, Starc and Wollmann, 2024). Our analysis of repeated acquisitions within CPC technology classes provides complementary evidence in the digital sector and speaks directly to this evolving regulatory framework.

Third, we contribute to a broad literature on markets for technology (e.g., Arora and Gambardella, 2010, *inter alia*), that frequently uses patent data to measure innovation outcomes. The use of Cooperative Patent Classification (CPC) groups as a proxy for technological domains follows prior work such as Hall, Jaffe and Trajtenberg (2001) and Bloom, Schankerman and Van Reenen (2013), who relied on alternative (coarser) patent classification schemes. We also draw on insights from the literature on patent citations as a measure of knowledge spillovers (Jaffe, Trajtenberg and Henderson, 1993).

Finally, our paper is related to the empirical literature on mergers and acquisitions, which has long documented that merger activity clusters in waves, by industry and over time (Andrade, Mitchell and Stafford, 2001). The acquisition sequences we study are a finer-grained form of this pattern, waves within narrow technology areas, often driven by the same acquirer. Within this literature, prior work examines acquisitions as a mechanism for sourcing innovation, showing that active acquisition markets can increase innovation incentives, particularly for smaller firms that can monetize successful innovation through acquisition (Phillips and Zhdanov, 2013). Similarly, acquiring firms often innovate more after acquiring innovative targets and recombining technologies and inventor capabilities (Bena and Li, 2014; Li and Wang, 2023; Sevilir, Tian and Zhang, 2025). Our results complement this literature by tracing how innovation evolves around acquisitions beyond the acquiring firm itself, across the broader technological ecosystem, through citation and employee-mobility dynamics.

The rest of the paper is organized as follows. Section 2 presents the stylized model and its predictions. Section 3 describes our data sources and the construction of the CPC-year panel. Section 4 provides descriptive statistics on acquisitions, targets, and patenting patterns. Section 5 presents our main empirical analysis of post-acquisition patenting outcomes and

uses citation-based metrics to capture the influence and quality of innovation. Section 6 concludes with a discussion of implications for competition policy and future research.

2 A Model of Acquisition Waves

This section develops a stylized model of startup acquisitions in a technology space with uncertain commercial viability. Beliefs about viability evolve through Bayesian updating on public signals, and acquisitions and innovation co-move with beliefs. The model is deliberately compact. We sketch its building blocks here and relegate the full setup, derivations, and proofs to Appendix A.

The model generates several predictions. First, acquisitions cluster into waves: beliefs move persistently rather than resetting each period, so spells of high belief tend to generate several qualifying deals in succession, even though completing a deal itself weakly discourages the next one by crowding the space. Second, innovation activity rises before any acquisition in spaces that go on to be acquired, a selection effect across technology states rather than a causal consequence of the deal. Third, every acquisition durably raises the entry threshold relative to what it would have been absent the acquisitions, evaluated at the same belief. This crowding effect persists even if and when a wave stalls, though whether a wave stalls at all depends on the shape of the underlying primitives and is not guaranteed by our maintained assumption alone. This is a counterfactual comparison rather than a claim that the realized level of innovation must fall below its own pre-wave level, since beliefs continue to evolve independently of the acquisitions themselves. Fourth, when preventing standalone entry is worth more to the incumbent than the entrant’s own standalone value, the model produces *killer acquisitions* (i.e., deals in which the incumbent shelves rather than commercializes the target’s project). The first three predictions line up with the empirical patterns we document below. The fourth we cannot test directly because commercial success in our setting is much more difficult to track than in previously studied industries such as pharmaceuticals.

The model’s technology space corresponds empirically to a narrow patent technology class, the CPC main group of Section 3. Thus, the model predictions are statements about patenting and acquisition activity within a CPC group and are intended to map onto the empirical analysis. The clustering of deals corresponds to the descriptive patterns of Section 4.3 and the pre-acquisition rise and the persistent effect of accumulated acquisitions match the event studies of Section 5.1.

2.1 Setup

A single technology space hosts one incumbent platform firm and a sequence of short-lived potential entrants. The space has an unknown commercial viability state $\theta \in \{H, L\}$, and all participants update a common belief $\mu_t = \Pr(\theta = H \mid \mathcal{H}_t)$ as public signals $s_t \in \{h, \ell\}$ arrive, with an h -signal more likely under $\theta = H$ than under $\theta = L$. Belief drift is what links innovation and acquisition activity over time, since both the entrant's willingness to develop and the incumbent's willingness to acquire move in the same direction with μ_t .

Each period a potential entrant draws a project quality $x_t \geq 0$ from a distribution F that is independent of θ , then decides whether to pay development cost $c_E > 0$. If the project is developed, the incumbent may acquire it, and conditional on acquiring may either commercialize it (paying an irreversible cost $c_I > 0$) or shelve it. The number of prior entrants acquired by the platform at period t is denoted by N_t .

Three primitives summarize how acquisitions reshape the value of the technology space. The first is the *commercialization advantage* $\kappa(N_t)$, capturing complementary assets like distribution, integration, and data that make any given project worth more in the incumbent's hands than as a standalone product. The second is *competitive crowding* $\gamma(N_t)$, capturing how much of the space an independent entrant can still capture on its own when the incumbent has accumulated N_t prior acquisitions. The third is the *competitive externality* $\psi(N_t)$, the damage standalone entry inflicts on the incumbent's existing portfolio. Both γ and ψ depend on N_t because each prior acquisition extends the incumbent's footprint in the space. We maintain $\kappa(N_t) > 0$ and $\gamma(N_t) > 0$ for all $N_t \geq 0$ throughout, so that the entry and commercialization cutoffs introduced below are always finite and well-defined. Project values scale jointly with μ_t and x_t . The incumbent earns $\mu_t \kappa(N_t) x_t$ from commercialization, the entrant earns $\mu_t \gamma(N_t) x_t$ from standalone entry, and standalone entry costs the incumbent $\mu_t \psi(N_t) x_t$ in lost rents.

The incumbent makes a take-it-or-leave-it offer at price $P_t = \mu_t \gamma(N_t) x_t$, exactly the entrant's standalone value, so the entrant's net payoff from developing is $\mu_t \gamma(N_t) x_t - c_E$ whether or not the deal happens. The entrant therefore develops if and only if

$$x_t \geq \underline{x}(\mu_t, N_t) \equiv \frac{c_E}{\mu_t \gamma(N_t)}. \quad (1)$$

Higher beliefs lower the threshold and encourage entry, while the incumbent's accumulated presence erodes γ and raises the threshold over time.

The incumbent's acquisition decision partitions the parameter space (κ, γ, ψ) into three regimes. In the *commercialization regime*, the incumbent's complementary assets make the deal worthwhile, and it acquires (and develops) only projects of sufficiently high quality, those

with $x_t \geq x^* \equiv c_I / \{\mu_t[\kappa(N_t) + \psi(N_t) - \gamma(N_t)]\}$. This is the canonical buy-to-commercialize motive. In the *pre-emption regime*, defined by $\psi(N_t) > \gamma(N_t)$, defending the existing portfolio is worth more than the entrant earns on its own. The incumbent then acquires every developed project, commercializes those with $x_t \geq \bar{x} \equiv c_I / [\mu_t \kappa(N_t)]$, and shelves the rest as *killer acquisitions*. This is the only regime in which the model generates deals whose purpose is to suppress rather than capture innovation. In the *no-acquisition regime*, neither motive is strong enough to support a deal, and the entrant keeps the standalone value. Figure 1 collects the timing within a period and shows how the realized project quality determines the outcome in each of these three regimes.

We further assume that the combined surplus $\kappa(N) + \psi(N) - \gamma(N)$ is decreasing in N (Assumption 1). The intuition is an Arrow replacement effect, where each additional acquisition cannibalizes the rents the incumbent already earns from earlier deals in the same space and shrinks the marginal value of κ , possibly augmented by saturation of the incumbent's competitive moat that would cause smaller changes in γ . Whatever the underlying channel, each acquisition shrinks the surplus available for the next. This alone does not guarantee that a wave eventually stalls, since a declining combined surplus can remain bounded away from the no-acquisition boundary. Whether and when the surplus reaches that boundary depends on the shape of κ , γ , and ψ , so we treat the stalling of a wave as a conditional event throughout rather than one implied by Assumption 1 alone.

2.2 Results

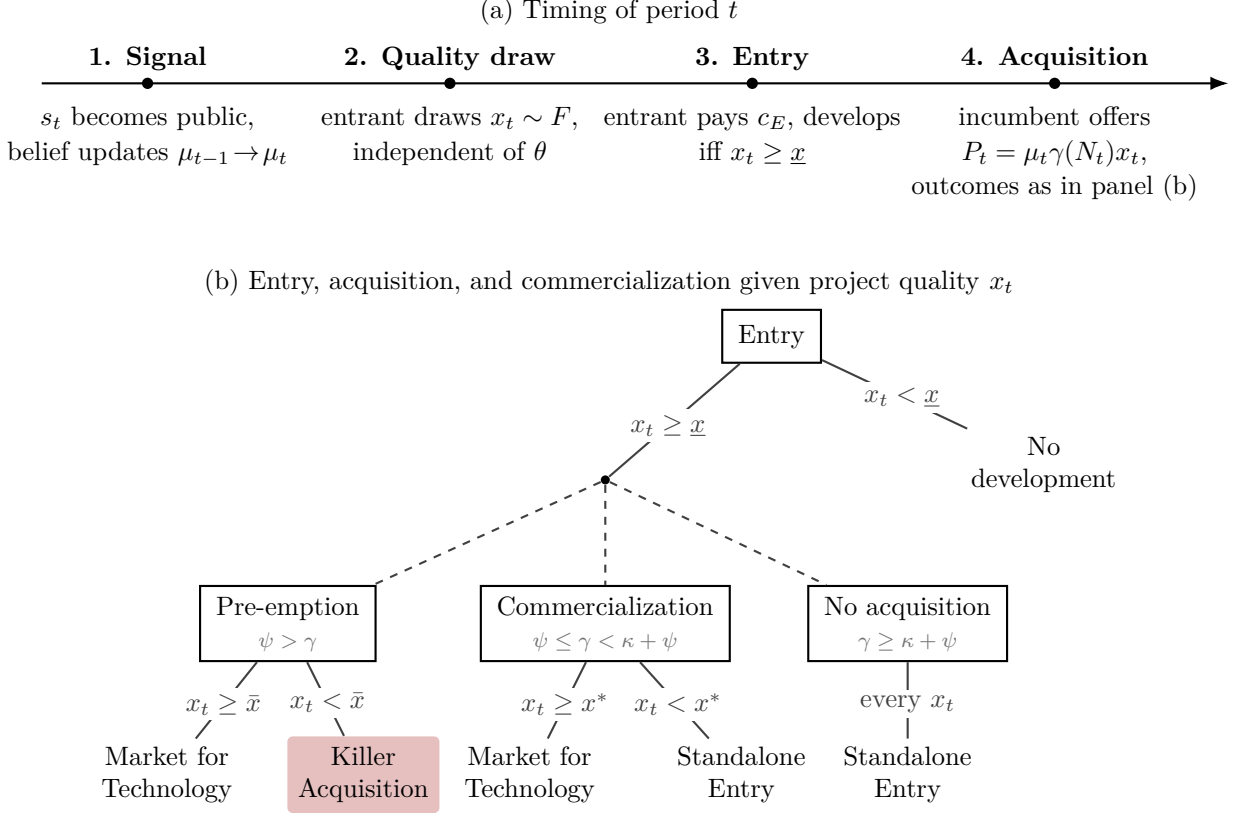
We formalize the main predictions of the model in three propositions, which we state with some short discussion below. We derive and prove them in Appendix A.

Proposition 1 (Belief persistence and pre-acquisition selection). *Let $a_t \in \{0, 1\}$ be an indicator that equals one iff an entrant is acquired in period t :*

- (i) *For fixed (N_t, θ) , $Pr(a_t = 1 \mid \mu_{t-1})$ is weakly increasing in μ_{t-1} , strictly so whenever an acquisition regime is active at N_t .*
- (ii) *Conditional on an acquisition occurring at date τ , $Pr(\theta = H \mid a_\tau = 1) > Pr(\theta = H)$, and for every $t < \tau$ the acquisition probability at t , conditional on $a_\tau = 1$, exceeds its unconditional counterpart.*

Part (i) reflects persistence in the belief process rather than a causal feedback running through completed deals. Because μ_t moves gradually rather than resetting each period, spells of high belief tend to produce several qualifying draws in succession, which is what

Figure 1: Timing and outcomes within a period



Notes: Panel (a) shows the four stages of period t , from the public signal that updates the belief μ_t to the incumbent's acquisition decision. Panel (b) shows the outcome as a function of project quality x_t , where the dashed branches are the three parameter regimes rather than choices.

generates clustering in event time. This is distinct from, and indeed works against, a channel in which the acquisition itself makes the next one more likely. Completing a deal raises N_t , which weakly *raises* the acquisition threshold through γ and, under Assumption 1, weakly lowers the surplus available for the next acquisition. Clustering here is a property of the belief path that acquisitions happen to track, not a positive feedback loop running through the transactions themselves. Part (ii) combines two forces: spaces that experience an acquisition are disproportionately high-viability spaces, and in high-viability spaces beliefs, and with them development rates, drift upward on average. Together they imply that acquired spaces will typically have seen rising development beforehand. In other words, the model predicts that the CPC-year event study models in Section 5.1 will find an innovation pre-trend.

After a wave ends, the incumbent's accumulated presence does not vanish. Because γ is weakly decreasing in N , the entry threshold remains elevated even once acquisitions cease, and the result is a permanent moat that delivers the entry-threshold prediction above. The

passive channel does not require any strategic suppression of entry by the incumbent, since its accumulated footprint alone makes independent commercialization harder than it would have been absent any acquisitions.

Proposition 2 (Post-wave moat). *The entry threshold $\underline{x}(\mu_t, N_t) = c_E/[\mu_t\gamma(N_t)]$ is weakly increasing in N_t . In particular, for any N_t with $\gamma(N_t) < 1$, $\underline{x}(\mu_t, N_t) > c_E/\mu_t$, so development is harder than it would have been absent any acquisitions. Once acquisitions cease at T^* , N_t freezes at N_{T^*} and the elevated threshold $c_E/[\mu_t\gamma(N_{T^*})]$ persists indefinitely.*

A final implication, internal to the wave and not tested directly in the data, is that whenever standalone entry is sufficiently damaging to the incumbent the model produces killer acquisitions. The economic logic is that once $\psi > \gamma$, the value the incumbent loses from a standalone competitor exceeds the value the entrant captures on its own, so the incumbent finds it worth paying to remove every developed project from the market, including those it never intends to commercialize itself.

Proposition 3 (Pre-emption and killer acquisitions). *In any period that falls in the pre-emption regime, defined by $\psi(N_t) > \gamma(N_t)$, the following hold.*

- (i) *The incumbent acquires every developed project.*
- (ii) *Projects with $x_t < c_I/[\mu_t\kappa(N_t)]$ are shelved rather than commercialized. These are killer acquisitions.*
- (iii) *If $\kappa(N_t)$ is decreasing in N_t , the commercialization threshold $\bar{x}(\mu_t, N_t) \equiv c_I/[\mu_t\kappa(N_t)]$ is increasing in N_t , so a larger range of developed-project qualities is shelved rather than commercialized.⁴*

The propositions describe two distinct senses in which technology spaces dominated by the incumbent become inhospitable to standalone innovation. One is passive, operating through accumulated competitive crowding (Proposition 2). The other is active, operating through pre-emptive acquisition followed by shelving (Proposition 3). The passive channel arises in every regime, while the active channel is internal to the pre-emption phase when it occurs.

This stylized model rationalizes a set of empirical facts about tech-sector acquisition waves that we document below. One strong simplifying assumption is that there is a single

⁴This is a claim about the commercialization cutoff itself, not about the *conditional share* of acquired projects that are killer acquisitions, $[F(\bar{x}_t) - F(\underline{x}_t)]/[1 - F(\underline{x}_t)]$. Because the entry threshold $\underline{x}_t = c_E/[\mu_t\gamma(N_t)]$ that defines the conditioning set also rises with N_t whenever γ declines, a rising $\bar{x}_t$ need not raise this share because this additionally requires restricting how fast κ falls relative to γ , or the shape of F , neither of which we assume.

acquirer (which allows us to characterize *ex post* competition in terms of a scalar state-variable, N_t). Introducing competition to acquire startups is unlikely to alter the “wave” dynamics, which are primarily driven by the exogenous drift of acquirer beliefs about commercial viability. Adding more potential acquirers does, however, seem likely to increase the probability of entry (by raising the expected acquisition price) and reduce the likelihood of killer acquisitions (since business stealing might come from incumbents in addition to entrants).

3 Data

We construct a novel dataset that combines acquisition activity with detailed patent-level information as well as employment trajectories of target and acquirer employees. This section describes our data sources, the construction of key variables, and the process we use to merge acquisition records with patent and employment data.

3.1 Acquisition Data

We begin by assembling a manually curated dataset of startup acquisitions by eight prominent technology companies, namely Amazon, Apple, Cisco, Facebook, Google, Intel, Microsoft, and Qualcomm. We selected these firms due to their longstanding presence in the digital and information technology sectors, their high frequency of acquisition activity, and their central role in current debates on digital competition and innovation policy. While Amazon, Apple, Facebook, Google, and Microsoft are typically characterized as digital platforms serving end users directly, firms like Intel, Cisco, and Qualcomm are equally important players, operating as key infrastructure and input providers in the broader digital ecosystem. These firms specialize in semiconductors, networking hardware, and communication protocols, and their acquisitions often target upstream innovations essential for enabling broader technological progress.

All eight firms have drawn antitrust scrutiny for their acquisition behavior. Notable examples include the FTC’s investigation of Facebook’s acquisitions of Instagram and WhatsApp,⁵ the U.S. Department of Justice’s (DOJ) antitrust lawsuit against Apple alleging monopolistic practices in the smartphone market,⁶ and the FTC’s inquiries into Microsoft’s,

⁵Federal Trade Commission, “FTC Sues Facebook for Illegal Monopolization,” December 9, 2020, <https://www.ftc.gov/news-events/news/press-releases/2020/12/ftc-sues-facebook-illegal-monopolization>.

⁶U.S. Department of Justice, “Justice Department Sues Apple for Monopolizing Smartphone Markets,” March 21, 2024, <https://www.justice.gov/archives/opa/gallery/justice-department-sues-apple>.

Google’s and Amazon’s investments in generative AI companies like OpenAI and Anthropic.⁷ Qualcomm’s proposed acquisition of NXP Semiconductors faced global regulatory delays before collapsing in 2018,⁸ while Intel and Cisco have also faced merger reviews due to their active roles in consolidating upstream technology markets. In addition, the European Union’s Digital Markets Act (DMA), obliges designated “gatekeepers” (including all five GAFAM firms) to notify the European Commission of any planned acquisition involving a digital sector company, even if the deal does not meet the formal merger notification thresholds at the EU or national level.⁹ Our data capture this diverse spectrum of acquisition strategies and regulatory attention across platform and infrastructure firms alike.

For each acquisition, we record the name of the acquired company, the year in which the acquisition occurred, and, where available, information about the target’s deal value. Research assistants compiled the list from public reporting on each acquirer’s transactions; it contains 1,209 deals announced between 1987 and 2020. In order to focus on startup acquisitions, which we define as acquisitions involving relatively small and early-stage companies, we exclude large deals involving targets with extensive patent portfolios (e.g., Apple’s acquisition of Intel’s modem business). These exclusions allow us to avoid confounding effects from horizontal consolidation and instead focus on the acquisition of emerging innovators. We identify these deals by inspection of the target names rather than by a mechanical cutoff, and Table B.3 in the appendix lists all 19 of them together with the number of patents the target had filed before the acquisition. These range from 90 to 42,876 patents, against a median of zero and a maximum of 504 among the deals we retain. This leaves 1,190 acquisitions of emerging innovators, comprising the vast majority of all acquisitions by the eight technology companies under study.

3.2 Patent Data

To characterize technological activity, we rely on data from the PatentsView database (USPTO 2016). Specifically, we extract data on patent filings and grants, including application years and Cooperative Patent Classification (CPC) codes. We also obtain information

monopolizing-smartphone-markets.

⁷Federal Trade Commission, “FTC Launches Inquiry into Generative AI Investments and Partnerships,” January 25, 2024, <https://www.ftc.gov/news-events/news/press-releases/2024/01/ftc-launches-inquiry-generative-ai-investments-partnerships>.

⁸Reuters, “Qualcomm Ends \$44 Billion NXP Bid After Failing to Win China Approval,” July 26, 2018, <https://www.reuters.com/article/technology/qualcomm-ends-44-billion-nxp-bid-after-failing-to-win-china-approval-idUSKBN1KF18X/>.

⁹Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on contestable and fair markets in the digital sector (Digital Markets Act), <https://eur-lex.europa.eu/eli/reg/2022/1925/oj>.

on patent assignees, which allows us to link patents to the firm that holds the right at the time of issuance.

Throughout the paper, we use CPC groups to group together patents in related technology areas. The CPC system is a hierarchical system devised by the US Patent and Trademark Office (USPTO) and the European Patent Office (EPO). The main purpose of the CPC system is to facilitate prior art searches by patent examiners, so the patents within a code correspond to similar technologies (though we would not go so far as to argue that they constitute technology markets in the sense of having a similar level of substitutability among patents). All US patents are assigned at least one primary CPC group, and possibly several secondary codes. The system has over 250,000 distinct categories, which allows for billions of possible combinations. Throughout this paper, we refer to the six-digit main group associated with a patent’s primary CPC group as that patent’s “CPC class.”

We link each acquired startup to its patent portfolio in two steps. The first is a rule-based match: target and assignee names are lowercased and stripped of a leading article, of punctuation, of 26 legal and industry suffixes, and of whitespace, and are then required to be exactly equal, with no fuzzy or substring matching. We run it both over granted-patent assignees and over pre-grant publication assignees, linking the latter to granted patents through the USPTO crosswalk. The second source matters because a patent filed by the target before the acquisition often issues afterwards with the acquirer as assignee: the granted record then no longer carries the target’s name, while the pre-grant record still does. Manual inspection of the resulting matches confirms they are highly reliable.

Exact matching cannot resolve targets whose patents are filed under aliases, subsidiary names or division names. We therefore supplement the rule-based match with a hand-curated match between target names and PatentsView assignee organisations, in which ambiguous cases were resolved by inspection. Together the two yield 8,070 pre-acquisition patents across 458 of the 1,190 acquisitions, of which 7,003 come from the rule-based match and 1,067 from the curated match alone. A patent enters a target’s portfolio only if its filing date precedes the acquisition date, so portfolios are pre-acquisition by construction; where a patent matches more than one target, we assign it to the earliest acquisition.

To focus our analysis on technology areas most relevant to the digital sector, we restrict attention to patents classified under CPC sections G (Physics) and H (Electricity) with 1,289 CPC classes. These two sections encompass the majority of innovation activity in information technology, software, electronics, and telecommunications.¹⁰

¹⁰While this restriction excludes a small number of patents filed by acquirers or targets in our sample, it screens out a large number of CPC classes that are never used by firms in our sample, and it dramatically reduces computational burdens. Across the entire corpus of US patents, there are 7,005 CPC classes, and each class has an average 30.3 new patent applications per year.

3.3 Data on Employee Work Histories

We use data from Revelio Labs to analyze employee-level career trajectories before and after acquisition events. Revelio Labs aggregates publicly available resumes, primarily from LinkedIn, to construct standardized longitudinal records of individuals’ employment histories. These data allow us to observe job transitions, employer affiliations, and role durations at the individual level, making them well-suited to study labor market dynamics around acquisitions.

To link employees to acquired firms, we begin with our list of 1,190 startup acquisitions and search for exact matches between cleaned target firm names and employer names in the Revelio data. Name-based matching is considerably more error-prone here than in the patent data: nearly any company can have a LinkedIn presence, whereas patent assignees are limited to firms that file patents, so unrelated firms sharing a target’s name are far more common. We therefore reviewed every candidate match by hand, discarding those flagged as problematic, and validated additional matches recovered through name variants or patent-assignee evidence against Revelio’s founding year, web domain, and corporate-parent records.¹¹ Because Revelio data are sourced largely from LinkedIn, which was founded in 2003, and because we require pre-acquisition employment histories to evaluate changes over time, we limit the sample to acquisitions occurring in 2005 or later. This restriction yields 932 acquisitions. Of these, we manage to cleanly identify and verify 692 targets in the Revelio data. We match on cleaned names: target names are lowercased, punctuation is replaced by spaces while domains, hyphens and parentheses are preserved, and a list of common corporate terms is removed, after which the cleaned name must match a Revelio company record exactly. We similarly identify the acquiring firms in the Revelio data. In doing so, we map not only to the parent-acquiring entity but also include all affiliated subsidiaries. For instance, positions listed at Instagram, DeepMind, or GitHub are treated as positions at Facebook, Google, and Microsoft, respectively, after their respective acquisition dates.

Résumé coverage is not random, and the linked sample differs from the acquisitions we cannot match. Table B.5 compares the 692 matched targets with the 240 unmatched. Matched targets are far more likely to hold pre-acquisition patents (48 against 18 percent), hold about four times as many, are acquired roughly a year and a half later, and are considerably larger where deal values are disclosed. Much of this operates through patent holding itself: among targets that do hold patents, 88 percent are matched and the two groups’ portfolios are statistically indistinguishable (Panel B), so the linked sample is closer to the

¹¹Revelio Labs also offers matched patent data, which we use to check and enhance the matching of patent-holding targets.

population entering the citation analysis than the overall comparison suggests. Differences nonetheless remain within that group, where matched targets are acquired about two years later and are larger. The workforce results therefore describe the larger and more recent part of our sample rather than the average acquisition, and the retention rates in Section 4.4 should be read accordingly.

When matching inventors, we restrict attention to inventors on pre-acquisition patents of acquired targets. Inventors are identified from PatentsView’s disambiguated inventor data, and we keep every inventor-patent pair for which the patent is in our analysis sample of target patents. For each inventor in the roster we link them to a Revelio user-id using Revelio’s curated patent-to-user matching shards, and consider inventors to be reliable matches if Revelio’s position data record them as employed at the same target firm in any year up to and including the acquisition year. We are able to identify at least one reliable inventor match for roughly half of acquired patents.

The resulting data provide a dynamic view of workforce changes surrounding acquisition events, allowing us to estimate retention patterns, employee turnover, and the relationship between post-acquisition integration and innovation outcomes. This labor market lens complements our patent-based analyses and offers a more complete picture of the organizational consequences of digital acquisitions.

3.4 Dataset Construction

Our empirical analysis uses three panel datasets. In the first dataset, each observation represents a unique technological field (CPC class) in a specific year. For each CPC-year combination, we calculate the number of patent applications filed, identify whether an acquisition occurred involving a target firm that had previously patented in that CPC group, and record which of the eight focal technology firms carried out the acquisition. This structure allows us to examine changes in innovation activity (measured by patenting intensity) before and after acquisition events within the same technological area. In addition, it enables us to compare CPC groups with acquisitions to those that were never subject to one, providing the comparison group for the event-study analyses below.

We construct several variables to characterize the acquisition behavior of each firm and the structure of innovation within each CPC group. These include indicators for whether the acquirer had patented in the CPC group prior to the acquisition, whether the acquisition represented the first such transaction in that technology area, and whether multiple acquisitions occurred in the same CPC group over time. These variables allow us to explore patterns of repeated acquisitions, the role of technological expertise, and the persistence of

innovation dynamics following a merger.

In the second panel, each observation represents a single patent in a given year. We restrict the sample to patents in CPC groups containing at least one patent filed by a target firm before its acquisition, and track whether each patent was filed by the acquired firm or by a third party. A patent counts as acquired only if its filing date precedes the acquisition date, so the treated set is the technology the target had developed at the time of the deal and excludes anything the target filed afterwards. Applications filed before the acquisition but granted after it are included: they account for roughly a third of the acquired patents, and we identify them through pre-grant publications, since their granted records typically list the acquirer rather than the target as assignee. Our main outcome variable is the number of prior-art citations that a focal patent receives from other U.S. patents filed or granted in a particular year. These forward citations are a widely used measure of the economic and technological significance of the cited patent, and the panel structure allow us to document the relative importance of specific patents through time, before and after each acquisition event.

In the third panel, each observation represents an individual inventor in a given year. The sample covers inventors named on a target’s pre-acquisition patents whom we can reliably link to a Revelio work history, observed over a window of five years on either side of the acquisition. For each inventor-year we count patent filings across all assignees, distinguishing filings at the acquirer from filings elsewhere. We also classify each inventor’s employment status one year after the acquisition, distinguishing inventors who stayed at the acquirer from those who left, and, among the latter, those who had already left the target before the deal closed. This panel underlies the inventor-level analysis.

These three datasets provide a foundation for analyzing how acquisitions influence innovation in targeted technology areas. By combining patent data with manually collected acquisition records, we are able to track the timing, intensity, and nature of innovation before and after each acquisition event, at the broad technology level, the level of the individual patent, and individual inventors. This structure enables us to evaluate the relationship between acquisition activity and technological development with a relatively high degree of granularity.

4 Descriptive Statistics

4.1 Big Tech Acquisitions

Table 1 provides an overview of the acquisition activity by each of the eight major digital firms included in our analysis, along with summary statistics on the patent holdings of their targets. The first column shows the number of acquisitions for each acquirer in our sample. The second shows the share of targets with at least one patent, and the following columns report the average size and technological breadth of the patenting targets’ portfolios, both overall and within CPC sections G (Physics) and H (Electricity), which account for the bulk of digital technology innovation. Because each CPC group in which a target had patented before its acquisition defines an event in the event studies below, one acquisition of a patent-holding target typically contributes four to five CPC-level events.

Comparing the various acquirers several patterns emerge. First, across all acquirers, a substantial share of startup targets do not hold any patents at the time of acquisition. The probability that an acquired firm holds at least one patent ranges from as low as 20% for Facebook to as high as 80% for Qualcomm. This heterogeneity likely reflects differences in acquisition strategies, with some firms (particularly those in semiconductors and hardware-intensive sectors) placing greater emphasis on acquiring patented technologies. Second, conditional on acquiring a target with patents, the average number of patents also varies across firms. Amazon, Intel and Qualcomm stand out with average target portfolios exceeding 20 patents per acquisition, and Intel’s and Qualcomm’s targets also span the widest technological scope, at more than seven distinct CPC groups each. In contrast, targets acquired by Facebook have both the smallest portfolios and the narrowest scope, five patents across 2.5 CPC groups. Amazon combines the two patterns: just 28% of its targets had applied for a patent prior to acquisition, but those that did owned 31 patents on average, spanning 5.3 CPC groups, suggesting that many of its deals target capabilities beyond formal intellectual property such as talent, products, or user data.

The last two columns of Table 1 report one-year retention among employees of the acquired firm. On average, 31 percent of target employees have left the acquirer within a year of the deal, and retention varies considerably across acquirers, from 55% at Apple to 76% at Intel. The final column shows the same retention rates for the target’s inventors: 77 percent on average, ranging from 70% at Google to 94% at Intel. Interestingly, inventors are retained at higher rates than the workforce as a whole at every acquirer except Microsoft, where the two rates are essentially equal, and the size of that gap varies considerably, from five percentage points at Google to twenty at Apple.

Table 1: Number of targets, patents acquired, and retention rates by each company

	#Acquisitions	Pr(AnyPatent)	#Patents	#CPC Groups	#CPC Groups (G&H)	Overall Retention (1y)	Inventor Retention (1y)
Amazon	103	0.28	30.55	5.31	4.93	0.65	0.72
Apple	126	0.41	17.65	4.69	4.31	0.55	0.75
Cisco	222	0.41	13.29	4.34	4.18	0.72	0.84
Facebook	102	0.20	5.15	2.50	2.25	0.68	0.79
Google	260	0.33	14.87	3.46	3.17	0.65	0.70
Intel	97	0.64	25.23	7.23	6.97	0.76	0.94
Microsoft	236	0.35	15.92	4.11	3.90	0.74	0.72
Qualcomm	44	0.80	22.51	8.06	7.80	0.69	0.79

Table 2: Median deal values (in millions of USD)

	Target has no patents	Target has patents
Amazon	96.50	275.00
Apple	25.00	200.00
Cisco	124.50	265.50
Facebook	40.00	200.00
Google	70.00	191.00
Intel	180.00	207.50
Microsoft	127.50	200.00
Qualcomm	25.00	157.00
Total	100.00	202.00

Overall, the data in Table 1 highlight the diversity of acquisition patterns across large digital firms and underscore the importance of distinguishing between acquisitions of patent-rich and patent-poor targets. These patterns also motivate our event-study approach, which tracks innovation outcomes in narrowly defined technological fields before and after acquisition events.

Table 2 reports median deal values (in millions of USD) for acquisitions by each of the eight focal technology firms, separately for targets with and without patent holdings.¹² The results reveal a clear pattern. Across nearly all firms, acquisitions of patent-holding targets are associated with substantially larger transaction values. These findings reinforce the importance of distinguishing between acquisitions that involve patented innovation and those that do not. They also suggest that while many acquisitions are motivated by access to capabilities or personnel, patents remain a key driver of valuation and strategic interest for big tech acquirers.

For instance, Apple paid eight times more for the median patent-holding target (\$200 million) than for the median target without patents (\$25 million). Similarly, Google’s median deal size increases from \$70 million to \$191 million when the target holds patents. This suggests that patent holdings may serve as a signal of technological quality, maturity, or strategic value, especially in software-oriented acquisitions. The size of the premium varies widely across acquirers, from 15 percent at Intel to more than sixfold at Qualcomm and eightfold at Apple. This patent premium is similarly large when taking the size of the target into account.¹³

¹²We focus on median rather than mean deal value because the latter statistics is sensitive to outliers.

¹³In Table B.4 in Appendix B.5, we show that deal values are much larger when patents are present on

Figure 2: Acquisition Time Trends

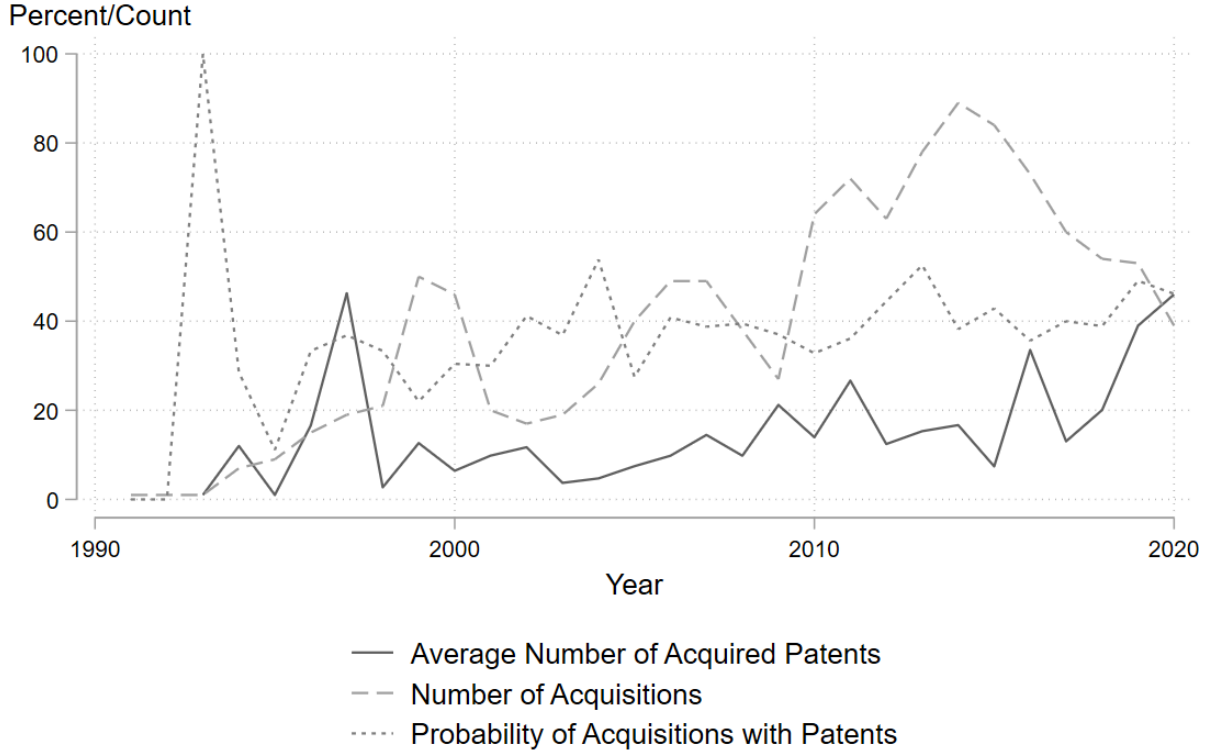
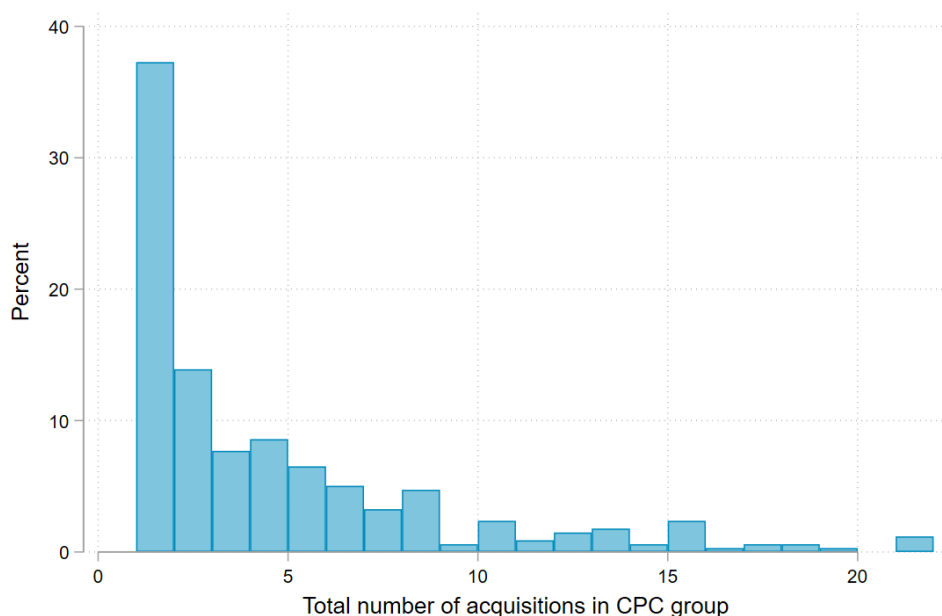


Figure 2 illustrates the annual frequency of startup acquisitions over the sample period. The figure serves two key purposes. First, it documents the overall volume and dynamics of acquisition activity across years. Second, it provides visual context for understanding the timing of acquisition events in our event-study framework.

The data show that acquisition activity is unevenly distributed over time, with noticeable peaks in certain years. These surges often coincide with broader trends in the technology sector, such as funding booms, IPO droughts, or platform expansion phases. For example, there is a pronounced increase in acquisition activity during the early 2010s, which corresponds to a period of aggressive growth by firms like Google, Facebook, and Amazon. The observed variation in acquisition timing underscores the need to control for year fixed effects in our empirical specifications. It also highlights the importance of matching event and comparison technological areas based on pre-trends in innovation, as different cohorts of acquisitions may have been subject to distinct macroeconomic or regulatory conditions.

average, even after controlling for the number of target firm employees. In fact, having more employees seems to amplify the premium paid for targets with patents.

Figure 3: Distribution of the number of acquisitions across CPC groups with acquisitions



4.2 Acquired Patent CPC Groups

From this point forward, we restrict our analysis to 1,289 CPC groups within sections G (Physics) and H (Electricity). This restriction is analytically useful because it concentrates our attention on core technology areas most relevant to the digital economy, including electronics, telecommunications, and software.¹⁴ The restriction to sections G&H is particularly useful when we select un-treated comparison CPC groups, since many irrelevant CPC codes are never touched by an acquisition. In total, 338 distinct G&H CPC groups are affected by at least one acquisition in our sample. Among these CPC groups, approximately 37% experience only a single acquisition event, while the remaining 63% experience multiple acquisitions, reflecting repeated acquisition activity within the same technological field. This recurrence suggests that certain technology areas serve as ongoing targets of strategic interest for digital incumbents.

Figure 3 provides a histogram of the number of acquisition events in each CPC group, conditional on the group experiencing at least one acquisition. The distribution is highly skewed. While many CPC groups experience only one or two acquisition events, a non-negligible number of technology areas see repeated acquisition activity, with some codes linked to as many as ten acquired firms over the sample period. This pattern helps motivate

¹⁴As shown earlier in Table 1, this focus does not meaningfully reduce the breadth of our sample. The average number of G&H CPC groups affected per acquisition is very similar to the average across all CPC groups, indicating that most patent-relevant startup acquisitions already fall within these sections.

Table 3: Previous patent experience for Acquirer’s first acquisition in CPC

	#Events	Any Patents before Acq	Share of CPC group patents	Min	Max
Amazon	85	75	0.0061	0.0001	0.0319
Apple	124	111	0.0071	0.0001	0.0349
Cisco	113	92	0.0089	0.0001	0.0767
Facebook	31	24	0.0050	0.0001	0.0292
Google	132	108	0.0056	0.0000	0.0362
Intel	190	177	0.0180	0.0007	0.1055
Microsoft	117	73	0.0034	0.0001	0.0431
Qualcomm	143	130	0.0162	0.0001	0.1210

Notes: An *event* is a (CPC main group, year) pair in which the given acquirer has made its first deal in given CPC group. Multiple deals in the same CPC-year count once. *Any Patents before Acq* count the number of events for which the acquirer already had a patent in that CPC group prior to the acquisitions. *Share of CPC groups patents* indicates the average share of acquirer patents among all patents in the CPC group at the time of the first acquisition.

our investigation of repeated acquisitions.

Table 3 examines the technological experience of each acquirer at the time of their first acquisition within a given CPC group. For each firm, we report the number of such first-time acquisition events, the number of cases in which the acquirer had previously patented in the same CPC group, and the share of total patents held by the acquirer within those groups.

The table shows that acquirers frequently enter a CPC group where they have a foundation of prior patenting activity. For example, Intel already owned patents in 177 out of the 190 CPC groups it entered through acquisition, and Qualcomm in 130 of 143. Even acquirers more typically associated with software and platform services acquire predominantly in domains where they already hold intellectual property: the share of first acquisitions preceded by the acquirer’s own patents in the group is 82 percent for Google and 77 percent for Facebook, and at its lowest, for Microsoft, still 62 percent.

The final three columns of Table 3 report the average share of patents in each CPC group that are held by the acquirer prior to the acquisition, along with the minimum and maximum values of this share across CPC groups. These shares are generally small in absolute terms, with average values below 2%, though the maximum reaches 11% for Intel and 12% for Qualcomm. Intel and Qualcomm also exhibit the highest average shares, suggesting more entrenched innovation activity within the technological domains they target. At the other end of the spectrum, Microsoft, Facebook, and Google tend to have lower pre-acquisition shares, consistent with a more exploratory or externally driven innovation strategy. Inter-

estingly, Microsoft seems to enter CPC groups where it has patented least often and, when it has, holds the smallest share, so its acquisitions are the least anchored in its own prior patenting.

4.3 Acquisition Waves

Given that many CPC groups experience multiple acquisitions, as predicted by the model in Section 2, it is informative to examine the structure and sequence of these acquisition waves. Table 4 documents the extent to which narrow technology areas are the site of repeated acquisition activity, both across and within acquirers. The unit of observation is an (acquirer, CPC main group, year) event, with multiple deals by the same acquirer in the same CPC-year collapsed into one.

For each of the eight acquirers, Table 4 reports the number of acquisition events, the share of events located in CPC groups with more than one deal-year, and the average number of years until the next deal in the same group. The latter two measures each come in two versions: one counts deals by any of the eight acquirers, the other only deals by the same acquirer. The wave shares are symmetric in time: an event is part of a wave if its CPC group sees deals in other years, whether before or after the focal event. The timing columns instead look forward to the next distinct deal-year, so several deals in the same year are not counted as zero-year gaps. Because the next deal by any acquirer arrives no later than the next deal by the same acquirer, “Years to Any Next” is weakly smaller than “Years to Own Next” at every event where both are defined. The two wave shares separate the sources of repeat activity: $\Pr(\text{Own Wave})$ captures the focal acquirer returning to the group, while the gap between $\Pr(\text{Wave})$ and $\Pr(\text{Own Wave})$ captures the other seven firms acquiring in the same area. The timing averages condition on a next deal of the relevant type existing, so they describe the pace of repeat activity rather than as a decomposition of it.

Several insights emerge. First, firms differ markedly in how often they return to a CPC group they have already touched. The share of events in a CPC group where the same acquirer deals in more than one year ranges from 0.47 for Facebook to 0.84 for Cisco, with Microsoft (0.77) and Intel (0.75) also high. Second, repeat activity in a CPC group often involves more than one acquirer. $\Pr(\text{Wave})$ exceeds $\Pr(\text{Own Wave})$ for every firm, and the gap is widest for Facebook (0.93 against 0.47) and Amazon (0.94 against 0.63), which acquire disproportionately in CPC groups that other acquirers also touch.

Figure 4 presents, side by side, histograms of the number of years between successive distinct deal-years within the same CPC group, conditional on a follow-on acquisition occurring. The left panel measures the gap to the next acquisition in the same CPC group by *any* of the

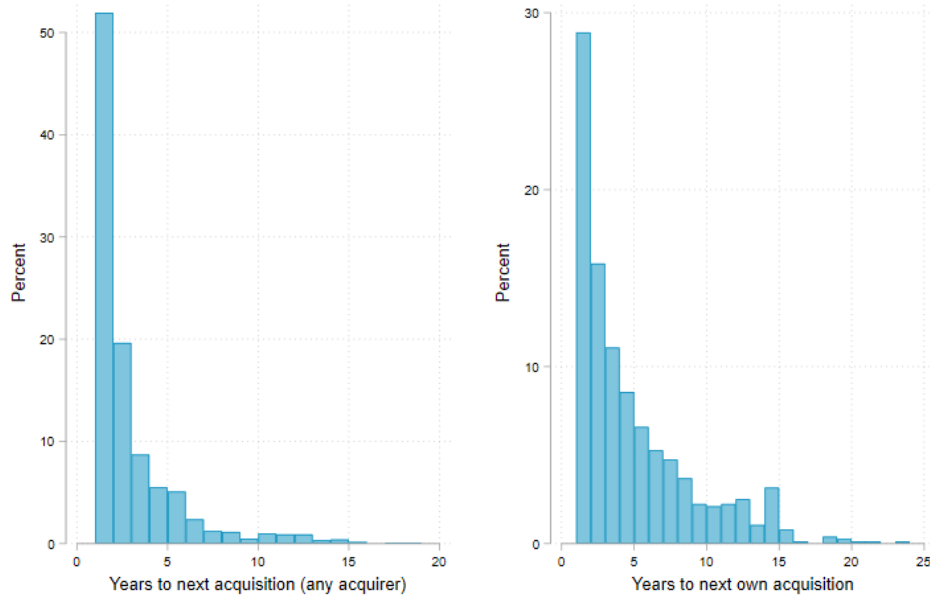
Table 4: Acquisition waves within CPC groups, by acquirer

	#Events	Pr (Wave)	Pr(Own Wave)	Years to Any Next	Years to Own Next
Amazon	142	0.94	0.63	1.75	4.07
Apple	214	0.93	0.65	1.90	3.99
Cisco	300	0.98	0.84	2.14	4.45
Facebook	45	0.93	0.47	1.58	2.43
Google	246	0.88	0.66	1.63	3.18
Intel	399	0.93	0.75	3.71	5.85
Microsoft	288	0.95	0.77	1.98	3.16
Qualcomm	261	0.91	0.70	2.25	3.19

Notes: An *event* is a (CPC main group, year) pair in which the given acquirer has at least one deal. Multiple deals in the same CPC-year count once. $Pr(Wave)$ is the share of the acquirer’s events in a CPC group with deals in more than one year (by any acquirer). $Pr(Own\ Wave)$ is the analogous share in which the same acquirer has deals in more than one year. $Years\ to\ (Any/Own)\ Next$ give the mean number of years to the next deal-year in the same CPC group, by any acquirer and by the same acquirer respectively. Same-year deals by other acquirers are not counted as zero-year gaps. Events with no subsequent deal-year of the relevant type are excluded from the mean.

eight acquirers. The right panel measures the gap to the same acquirer’s next acquisition in that CPC group. Same-year deals by different firms are not counted as zero-year gaps. Both panels record the time to the next *distinct* deal-year, so that the two measures are directly comparable. Events with no subsequent deal-year of the relevant type are dropped from the corresponding panel. Both distributions are highly skewed toward short intervals, indicating that CPC groups that see a follow-on acquisition tend to see it quickly. More than 50% of events in the any-acquirer panel are followed by another deal within one year, and more than 70% within two years. The corresponding shares in the own-acquirer panel are 29% and 44%. The own-acquirer distribution lies to the right of the any-acquirer distribution at every horizon: for each t , the share of events followed within t years is at least as high for any acquirer as for the same acquirer. This is what one would expect, since the any-acquirer gap is triggered by whichever of the eight firms returns first while the own-acquirer gap waits specifically for the same acquirer, though it is not guaranteed by the construction, because an event enters the any-acquirer panel whenever some acquirer returns and the own-acquirer panel only when the same one does. Restricting both panels to the 960 events where the same acquirer does return, 62.7% are followed by some acquirer within one year against 29.5% by the same acquirer, so on identical events other acquirers roughly double the pace at which a CPC group is re-touched. Overall, the short intervals in the left panel suggest that certain technological domains experience concentrated bursts of acquisition activity by multiple firms, consistent with heightened competitive interest or simultaneous recognition

Figure 4: Years to next acquisition within same CPC group



Notes: *Years to next (own) acquisition* is the number of years from an acquisition to the next distinct deal-year in the same CPC group, by any acquirer (left panel) and by the same acquirer (right panel), respectively. Because the measure is defined on deal-years, same-year deals by other acquirers are not counted as zero-year gaps. Acquisitions with no subsequent deal-year of the relevant type are undefined and excluded from the figure. Bars give the percentage of acquisition events at each gap length.

of strategic value. The longer intervals typical of the right panel indicate that individual firms' sustained acquisition activity in a given CPC group is a slower-moving process.¹⁵

Table 5 examines how an acquirer's prior patenting experience in a CPC group relates to the likelihood of follow-on acquisitions in that same technological domain. The results reveal a striking pattern. When the acquirer had already patented in the CPC group before the acquisition, 92% of first acquisitions are followed by additional acquisitions in that group. By contrast, when the acquirer had no prior experience in the CPC group, the first acquisition is much more likely to be an isolated event. This suggests that acquirers with established technological footholds are more likely to pursue sustained engagement in the same area, potentially reflecting greater absorptive capacity or strategic commitment. In contrast, firms entering unfamiliar domains may treat the acquisition as exploratory or one-off, with fewer follow-up investments.

¹⁵Figure B.5 in the appendix further explores the distribution of time between acquisition events. The gap measure is the same as in Figure 4 and Table 4, so same-year deals by other firms are not counted as zero-year gaps. Panel (a) truncates the range at six years. Panel (b) additionally restricts attention to acquisitions before 2015, so that every event has six full years in which a follow-on deal could be observed. The shape of the distribution is similar in both, with a large share of events clustered in the first two years.

Table 5: Follow-on Acquisition and Wave Positions by Experience

	#Events	No follow-on acquisition	First in acq wave	Last in acq wave	Mid in acq wave
No patents before Acq	145	0.42	0.36	0.06	0.16
Any patents before Acq	790	0.08	0.17	0.14	0.61

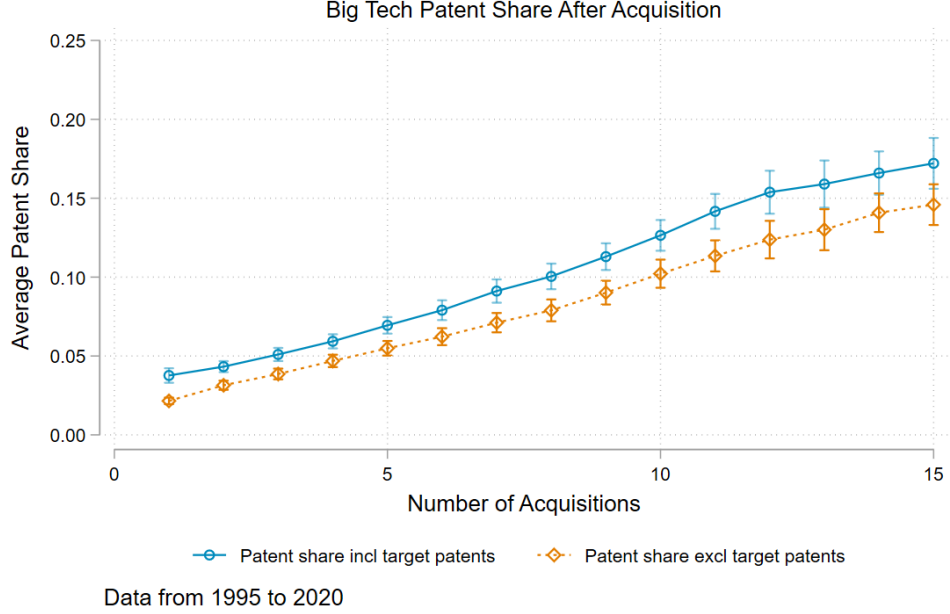
Notes: Each observation is an acquirer–CPC-group pair. The rows split these pairs by whether the acquirer already held patents of its own (i.e. non-target patents) in the CPC group by the year of its first acquisition there (*Any patents before Acq* vs. *No patents before Acq*).

Figure 5 shows how acquisition waves go together with more concentrated patent ownership within a CPC group. For each group, it plots the share of patents held by the eight large acquirers in the year after the k^{th} acquisition, against the cumulative number of acquisitions. The blue line counts all patents the acquirers own, including the ones they obtained from the targets. The orange line leaves out the acquired target patents, so it captures only the patents the acquirers filed themselves. Both shares rise steeply with the number of acquisitions, from about four to seventeen percent for the blue line and about two to fifteen percent for the orange line. CPC groups that see more acquisitions thus end up much more concentrated in the hands of big tech.

The two lines decompose this concentration into its accounting components. The gap between them is the share held through acquired patents themselves, the patents that move to the acquirer when it buys a target. This gap stays small, around two percentage points, and grows only a little as the number of acquisitions increases. The steep rise instead shows up in both lines. Even setting the acquired patents aside, the acquirers hold a far larger share of their own patents in exactly those CPC groups where they acquire the most. The concentration that goes together with acquisition activity thus consists mostly of the acquirers’ own patenting rather than of patents taken over from targets. This is an accounting statement rather than a causal one: as Table 3 documents, acquirers buy disproportionately in areas where they already patent, so the steep gradient partly reflects selection into CPC groups where acquirers were already active in patenting.

Acquisitions can nonetheless contribute to the acquirers’ own share. As the inventor evidence in Section 5.3 shows, acquired inventors who stay with the acquirer patent more after the acquisition, and the patents they file are assigned to the acquirer, so part of the orange line’s rise plausibly reflects the work of acquired inventors.

Figure 5: Big tech total patent share after acquisition



Notes: The figure plots the share of patents in a CPC group held collectively by the eight large acquirers at the group’s k -th acquisition, against the cumulative number of acquisitions k . The share is the acquirers’ patents divided by all patents in the CPC group, both counted as ten-year trailing sums. The blue line (*incl. target patents*) counts all patents the acquirers own, including those obtained from acquired targets; the orange line (*excl. target patents*) counts only the acquirers’ own patents, so the gap between the lines is the contribution of the acquired targets. Markers are means across CPC groups at each k , with 95% confidence intervals.

4.4 Employee Retention

Our final set of descriptive statistics concerns employee retention, and in particular how it differs between the target’s inventors and its wider workforce. The distinction matters for what follows: the citation analysis in Section 5.2 asks whether the acquired technology diffuses through the people who developed it, so what becomes of the inventors specifically, rather than of headcount in general, is the object of interest. For this analysis, we construct a panel of employee position-year observations running from the year before the acquisition, which is the year in which we identify the target’s workforce, through the fifth year after it.¹⁶ Job stints that begin and end in the same calendar year are excluded to reduce noise from short-term or uncertain employment records. In cases where a resume includes both a private-sector job and an academic or educational affiliation in the same year, we retain only the private-sector position. However, an individual may still hold multiple job positions

¹⁶To avoid double counting overlapping positions, we attribute the year of a transition to the new position rather than splitting the year.

across different firms in the same year if the data clearly indicate distinct roles.

Figure 6 plots retention for three groups, all identified in the year before the acquisition. *Target inventors* are employees of patent-holding targets whom we can confirm as inventors on the target’s pre-acquisition patents. *Target employees* are the full workforce of those same targets, inventors included, so the two series differ by the weight of the non-inventor majority rather than by the type of target. *Acquirer employees* are drawn from the acquiring firm and provide a baseline for ordinary turnover.

We construct the set of acquirer employees to have similar tenure profiles at the time of the deal. For each target employee in our sample, we identify an employee from the acquirer who began working there in the same year the target employee started at their organization. We focus on employees who work for the acquirer directly and exclude any subsidiaries that were targets in past acquisitions. If multiple acquirer employees meet this criterion, we select one at random to serve as the matched acquirer employee. This matching nets out differences in retention driven by career stage or job tenure. The baseline is matched on start year and employer but not on occupation, so it is not restricted to inventors; comparisons between the inventor series and this baseline therefore combine any acquisition effect with ordinary differences in turnover between technical and other staff.

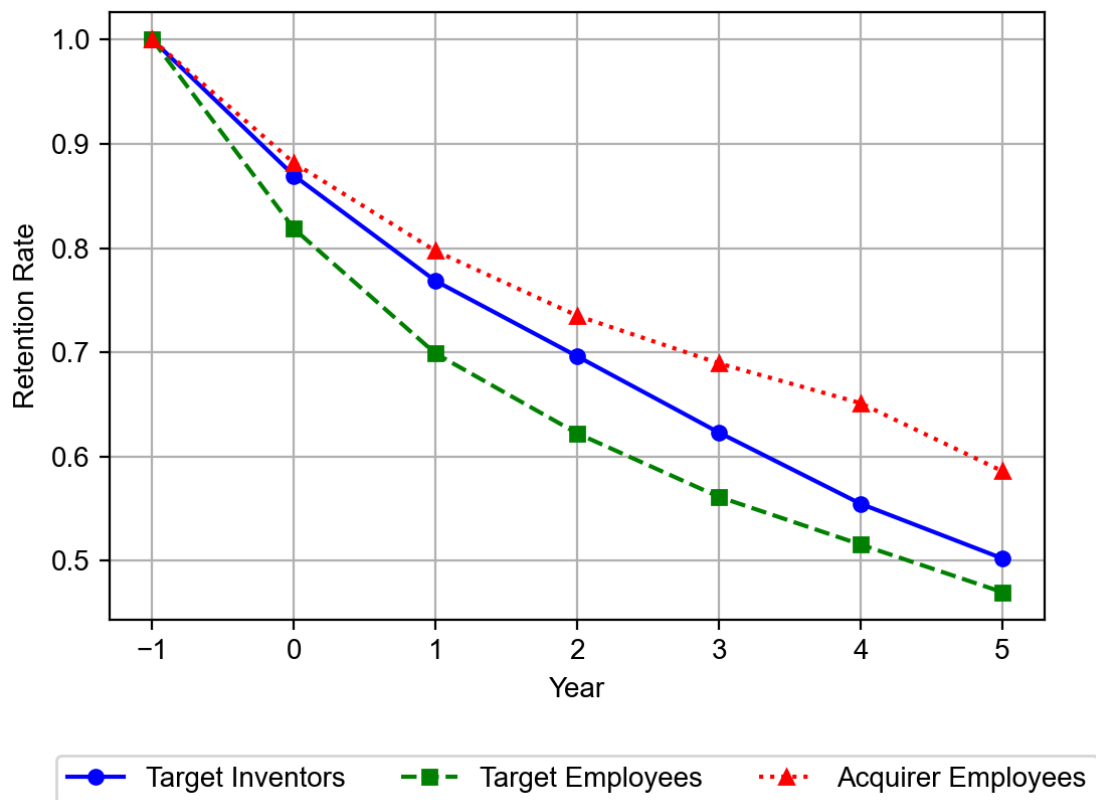
The horizontal axis shows years relative to the acquisition date, while the vertical axis measures the share of employees who remain at their original employer in each subsequent year. The figure reveals a stark divergence in retention dynamics following the acquisition. While retention among matched acquirer employees declines, retention among target employees falls much more sharply starting in the year of acquisition and continues to fall in subsequent years.

Specifically, weighting each employee equally, target employees experience a steep drop in retention in the first year after the deal: 31% have left within a year, and retention continues to decline thereafter. The fall is smaller among the target’s inventors, 23% of whom depart within the first year, so the technical staff continue at higher rates than the workforce as a whole. These are the one-year rates reported by acquirer in Table 1. In contrast, for most acquiring firms, the matched acquirer employees display a flatter retention curve, with substantially less attrition over the same time window.¹⁷

These patterns highlight the fragility of talent retention in the wake of technology (startup) acquisitions. Despite the strategic importance of acquiring human capital in tech-

¹⁷Figure B.8 in the appendix shows that this retention pattern is broadly consistent across all eight acquiring firms in our sample. While there is some variation in the level and slope of post-acquisition attrition (with Apple exhibiting the highest overall retention) every acquirer displays a substantial decline in target employee retention following the acquisition. However, for two firms, Cisco and Intel, attrition seems similar for both (acquired) target employees and directly acquired employees.

Figure 6: Retention Rates of Target Inventors, Target Employees, and Matched Acquirer Employees



nology deals, the evidence suggests that much of the target firm’s workforce departs within a few years of an acquisition.

5 Results

5.1 CPC Group Level Analysis

We first examine whether and how patenting activity within specific technological fields (defined by CPC groups) changes after an acquisition by a large digital firm. The CPC-by-year data allow us to observe innovation dynamics around acquisition events and assess whether these transactions coincide with shifts in the rate of innovation.

5.1.1 Baseline Event Study

The goal of this section is to describe how patenting activity within a technological domain evolves before and after acquisitions. By organizing the data at the CPC group–year level,

we can track how patenting trends change around an acquisition event, defined as the acquisition of a startup that had previously filed patents in that CPC group. We refer to CPC groups that experience at least one acquisition as *event groups* and to CPC groups that never experience one as *comparison groups*. Because the timing of acquisitions is itself an equilibrium outcome, and acquirers plausibly select technology areas based on their trajectories, we interpret the estimates below as descriptive evidence on the timing and shape of patenting changes around acquisitions rather than as causal effects of the acquisitions themselves. The pre-event coefficients are informative precisely on this point: they reveal whether acquisitions tend to occur in areas that were already on differential trends, which is what the model’s selection mechanism predicts.

Our primary outcome variable, $lpat_{ct}$, is the inverse hyperbolic sine, or arcsinh , of the total number of patents filed in CPC group c in year t . Our baseline specification estimates the evolution of patenting in a technological area around a startup acquisition:

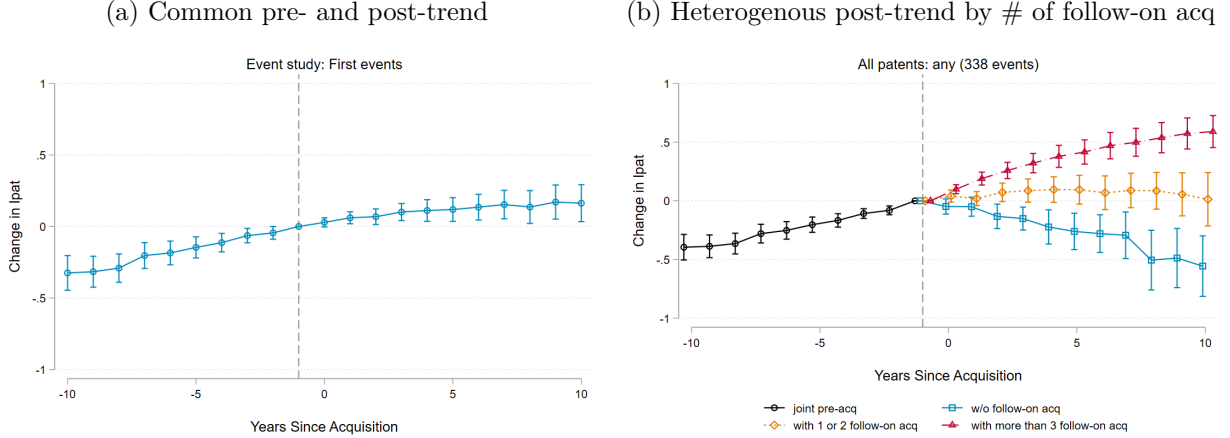
$$lpat_{ct} = \alpha + \gamma_c + \lambda_t + \sum_{s=-10}^{10} \tau_s \times d_{c,t=y+s} + \epsilon_{ct} \quad (2)$$

where γ_c are CPC group fixed effects and λ_t are calendar-year fixed effects. The indicator variable $d_{c,t=y+s}$ is set equal to one s years after a digital merger happened in calendar-year y in CPC category c . The coefficients τ_s measure the approximate percentage difference in patent filings s years from the acquisition, relative to the year before the event ($s = -1$), conditional on CPC group and calendar-year fixed effects. They thus summarize the within-CPC-group time path of patenting around acquisitions rather than a causal response to them.

Figure 3 shows that we often have multiple acquisitions within a CPC group. When two or more acquisitions occur in a CPC group in the *same year*, we treat them as a single event at the CPC-year level. This approach avoids double-counting and simplifies the structure of the event-study analysis while still capturing the cumulative footprint of overlapping acquisitions. It is less clear how we should handle multiple acquisitions in the same CPC group that occur in different years. We begin by focusing on the first event, with follow-on acquisitions providing a potential source of heterogeneity in post-acquisition dynamics. We then consider a specification that combines dummies for the first, second, and subsequent acquisition events.

The implicit comparison in equation 2 is to the comparison groups, which never experience an acquisition. Our event-study specification is therefore a hybrid: the coefficients τ_s are informed both by variation in the timing of acquisitions across event groups and by the

Figure 7: First acquisitions within CPC groups



Notes: Event-study estimates of patenting in a CPC group around its first acquisition. The outcome is the arcsinh of annual patent filings; the specification includes CPC-group and calendar-year fixed effects, with standard errors clustered by CPC group. Coefficients measure the approximate percentage difference in filings relative to the year before the acquisition ($s = -1$); caps indicate 95% confidence intervals.

contrast with comparison groups. Two-way fixed effects event studies of this form can be distorted when event timing varies across units, and robust estimators have been proposed by Callaway and Sant’Anna (2021) and Sun and Abraham (2021). Since our sample includes a large number of comparison groups, which anchor the counterfactual time path for CPC groups where acquisitions occurred, our simpler specification yields near-identical results to these estimators.¹⁸ Our results are also robust to alternative specifications that preserve the approximate percentage interpretation of the coefficients, such as Poisson quasi-maximum likelihood estimation, see Appendix B.1. Since CPC group-years with zero patents are rare, we use the more conventional arcsinh transformation in the main text.

Figure 7 presents the estimated event-study coefficients τ_s from equation (2), tracing total patenting within a CPC group around startup acquisitions.¹⁹ Panel (a) uses a pooled specification that dates event time from the first acquisition in each CPC group and estimates a single set of pre- and post-acquisition coefficients. The results show a marked upward trend in patenting before the acquisition, followed by continued growth afterward.

Panel (b) disaggregates the post-acquisition coefficients by the number of follow-on acquisitions that occur in the same CPC group, holding the pre-acquisition path common across groups. Patenting continues to grow after the first acquisition only where follow-on acquisitions occur, and the growth is stronger the more follow-on acquisitions there are.

¹⁸It might be interesting to explore how the estimates change in a conventional event study that uses only event groups, as discussed by Miller (2023).

¹⁹Graphs by acquirer can be found in Appendix B.5.

In CPC groups that see no further acquisition activity, patenting declines. Patenting thus rises most persistently in technological areas where digital incumbents continue to acquire patenting startups, a pattern consistent with ongoing strategic investment in the underlying technology.

Both panels also reveal strong pre-trends, with patenting activity already increasing in the years leading up to the first acquisition. Acquisitions are evidently not randomly assigned across CPC groups but occur disproportionately in areas with accelerating innovation. These pre-trends rule out reading the post-acquisition coefficients as causal effects, but they are precisely what the stylized model predicts. Because acquisitions concentrate in high-viability technology spaces, where beliefs and development rates drift upward, rising patenting ahead of an acquisition is what selection across technology states produces, rather than a causal effect of the deal itself.

The strong pre-trends in Figure 7 are direct evidence of selection: acquisitions tend to occur in CPC groups that are already experiencing rising innovation. To assess how much of the post-acquisition pattern reflects the trajectories of the areas acquirers select rather than the acquisitions themselves, we next implement a matching approach that pairs each event group with an observationally similar comparison group, requiring comparable levels and trends of pre-acquisition patenting. This comparison holds fixed the observable part of the selection and shows whether the post-acquisition patterns persist across more closely aligned trajectories.

To be clear, selection on unobservable characteristics of the CPC groups may remain even after matching. In particular, acquirers may have inside knowledge about future patenting activity in especially valuable CPC groups, leading them to acquire targets in areas that they know or expect will grow. An acceleration of patenting after acquisitions is therefore not by itself evidence that big tech acquisitions stimulate innovation.

5.1.2 Event Study with Matched Controls

The preceding analysis relied on all comparison groups. However, these CPC groups may differ systematically from event groups, especially in their pre-acquisition innovation trends. To address this concern, we construct a more closely matched set of comparison groups, pairing each event group with a single CPC group that experienced no acquisition but was on a similar patenting trajectory in the years immediately before the event.

Specifically, for each event group we measure two characteristics in the year before its first acquisition: the growth rate of patenting and the level of patenting, both on the arcsinh scale. We compare these to the same two characteristics of every CPC group without an

Table 6: Patent count and growth in CPC groups with acquisitions and without

	Event groups (1)	Comparison groups (2)	Difference (2) – (1) (3)	Matched comparison (4)	Difference (4) – (1) (5)
Patent growth, year before event (%)	9.14	0.52	-8.62***	8.82	-0.32
Patent count, year before event	217.98	26.05	-198.20***	199.37	-18.61
Observations	310	23,275	23,611	310	620

Difference columns are two-sample t -tests (unequal variances)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

acquisition, measured in the same calendar year, and select the comparison group that lies closest to the event group on the two dimensions jointly, measuring distance as the sum of the squared differences with each difference expressed in standard deviations of the corresponding characteristic. Standardizing in this way places growth and level on a comparable scale, so that neither dominates the match. The selected comparison group is assigned a pseudo-event year corresponding to its pair’s acquisition year. When several event groups are matched to the same comparison group, the comparison observation is duplicated and each copy inherits the respective event time. We retain only these matched pairs in the estimation sample. Matching requires a growth rate in the year before the acquisition; patent counts in our data begin in 1995, so this is unavailable for CPC groups first acquired in 1995 or 1996, which together with one CPC group that is discontinued mid-sample leaves 310 of the 338 event groups in the matched sample.

Table 6 reports summary statistics for event groups, for all comparison groups, and for the matched comparison groups. Both statistics are the characteristics the match conditions on, measured in the year before the acquisition. The goal is to assess how closely the matched comparison groups resemble event groups along these pre-acquisition dimensions. Column (1) shows the average patent growth rate and the number of patents in the year before the acquisition for event groups. Column (2) presents the same statistics for all comparison groups, while column (4) restricts attention to the matched subset. Columns (3) and (5) report the differences between event and comparison groups for the unmatched and matched samples, respectively.

The full set of comparison groups exhibits substantially lower growth rates and much smaller patent counts than the event groups, with large and statistically significant differences. The matched comparison groups, by contrast, closely resemble the event groups on both dimensions: the difference in pre-acquisition growth is 0.3 percentage points and the difference in patent counts is 19, neither statistically significant. The matching procedure therefore balances the pre-acquisition trajectory, so that the following comparisons are drawn

between similarly evolving CPC groups. We emphasize that this balance is achieved by conditioning on observed pre-acquisition patenting; it does not address selection on characteristics we do not observe.

After this matching procedure we estimate

$$lpat_{ct} = \alpha + \gamma_c + \lambda_t + \sum_{s=-10, s \neq -1}^{10} \nu_s \times d_{t=y+s} + \sum_{s=-10, s \neq -1}^{10} \tau_s \times d_{c,t=y+s} + \epsilon_{ct} \quad (3)$$

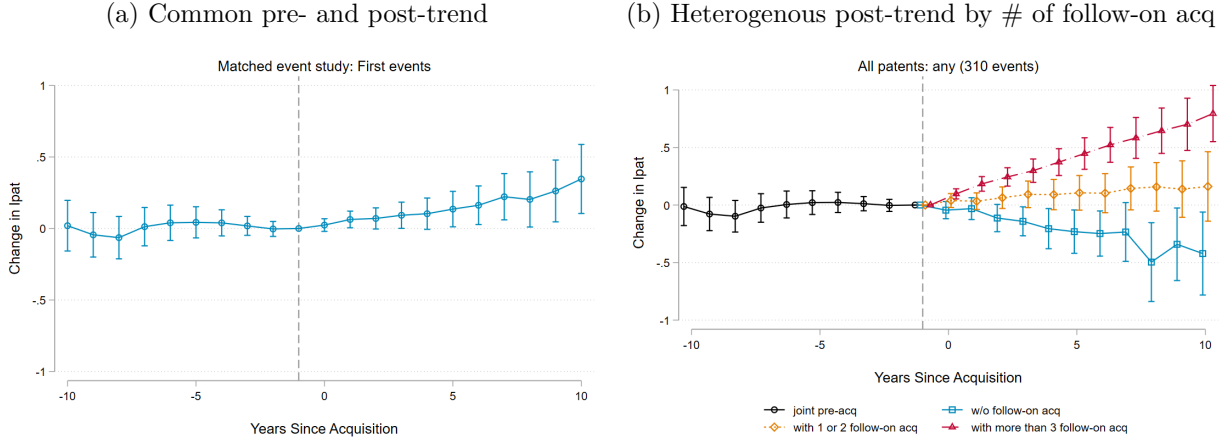
where γ_c are CPC group fixed effects and λ_t are calendar-year fixed effects. The indicator $d_{t=y+s}$ equals one for both the event group and its matched comparison group s years after the acquisition, with ν_s the corresponding coefficient, while $d_{c,t=y+s}$ equals one *only* for the event group c . The coefficients of interest are the τ_s : each measures how patenting in event groups differs from patenting in their matched comparison groups s years from the acquisition, relative to the year before the event. Because the match is constructed on the level of patenting in the year before the acquisition and on its growth over the preceding year, the coefficients at $s = -1$ and $s = -2$ are pinned down by construction: the first is the normalisation, the second is the matched growth difference restated. The coefficients at $s \leq -3$ are not targeted by the matching and therefore provide an independent check on whether the comparison groups were on a similar trajectory further back.

In line with the unmatched analysis, we examine how patenting in event groups evolves relative to their matched comparison groups, differentiating by whether and how many follow-on acquisitions occur in a given CPC group. Figure 8 plots the estimated coefficients τ_s . Panel (a) considers all first acquisition events and estimates a single set of post-acquisition coefficients, imposing a common path across event groups. Panel (b) relaxes this restriction and estimates separate post-acquisition paths by the number of follow-on acquisitions observed in each group.

To ensure that matching is based on comparable pre-acquisition dynamics, we require each event group to have at least two years of patent data before its first acquisition, which is what the growth rate entering the match is built from. Because patent counts in our data begin in 1995, this excludes the 27 CPC groups first acquired in 1995 or 1996, together with one CPC group whose patenting is discontinued mid-sample, leaving 310 of the 338 event groups.

The matched comparison eliminates the strong pre-trends of the unmatched analysis. Over the window the matching does not target, $s \leq -3$, we cannot reject that the coefficients are jointly zero ($F = 0.95$, $p = 0.47$), so the comparison groups track the event groups well beyond the two years the match conditions on. Patenting is persistent, so aligning two CPC

Figure 8: First acquisitions within CPC groups (matched)



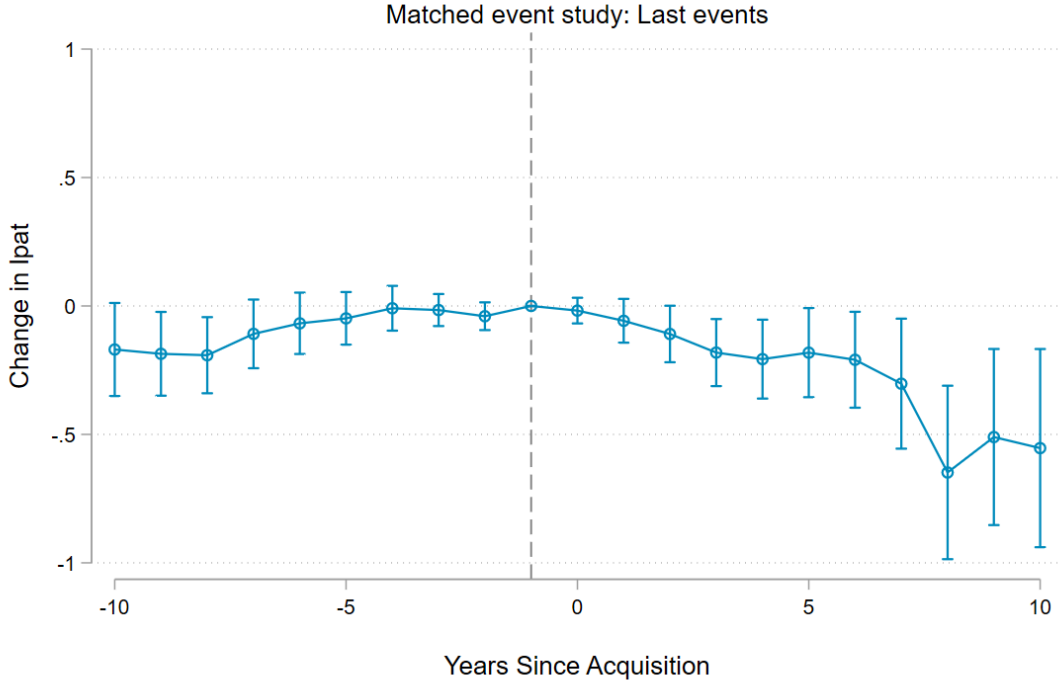
Notes: Matched event-study estimates of patenting in a CPC group around its first acquisition. The outcome is the arcsinh of annual patent filings. Each event group is paired with the CPC group without an acquisition that is closest on the growth rate and the level of patenting in the year before the event, both standardized. The plotted coefficients τ_s give the difference between event groups and their matched comparison groups by event time; the specification also includes event-time indicators common to both groups, CPC-group fixed effects, and calendar-year fixed effects, with standard errors clustered by CPC group. Coefficients measure the approximate percentage difference relative to the year before the acquisition ($s = -1$); caps indicate 95% confidence intervals.

groups in the year before the acquisition tends to align them for some years before that as well; these coefficients are therefore a check that the matching found genuinely similar groups rather than evidence that acquisitions are unrelated to a CPC group's prospects. After the acquisition the estimates show clear heterogeneity. CPC groups with several follow-on acquisitions show significantly larger increases in patenting, while CPC groups with a single acquisition show no statistically significant change in filings. This pattern is consistent with follow-on acquisitions accompanying more persistent investment and innovation activity in the targeted technology space.

The matched estimates in Figure 8 show how much the picture depends on whether further acquisitions follow. To see whether the pronounced rise in patenting in CPC groups with follow-on acquisitions precedes or follows those later deals, we re-estimate equation 3 centring event time on the *last acquisition* observed in a CPC group. Figure 9 shows no increase in patenting in event groups after the last acquisition, relative to their matched comparison groups.²⁰ Taken together with the rise around acquisitions that are followed by others, this suggests that patenting growth in these technology areas precedes the final deal

²⁰This is not an artefact of the matching: the same qualitative pattern appears in the unmatched event study centred on the last acquisition. Note also that the comparison groups here are those matched on patenting before each CPC group's *first* acquisition, so for groups with several acquisitions the pairing is anchored earlier than the event plotted.

Figure 9: Last acquisitions withing CPC groups (matched)



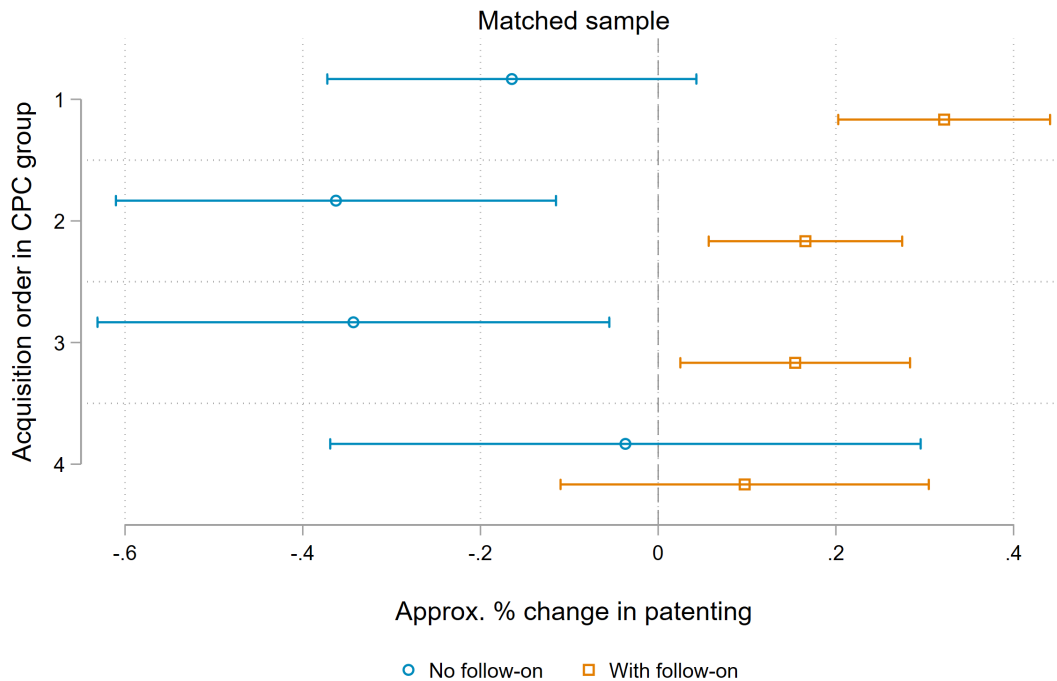
Notes: Matched event-study estimates of patenting in a CPC group around its last observed acquisition. The outcome is the arcsinh of annual patent filings. Each event group is paired with the CPC group without an acquisition that is closest on the growth rate and the level of patenting in the year before that group's *first* acquisition; event time is then measured from the last acquisition, so unlike the first-event figure the pairing is not constructed on patenting immediately before the event plotted here. The plotted coefficients give the difference between event groups and their matched comparison groups by event time; the specification also includes event-time indicators common to both groups, CPC-group fixed effects, and calendar-year fixed effects, with standard errors clustered by CPC group. Coefficients measure the approximate percentage difference relative to the year before the acquisition ($s = -1$); caps indicate 95% confidence intervals.

rather than responding to it, which is the selection pattern that part (ii) of Proposition 1 predicts.

To examine the timing of patenting growth relative to the sequence of acquisitions, we estimate a specification that compares patenting after the first, second, third and fourth acquisition in a CPC group, distinguishing in each case whether further acquisitions follow. Figure 10 reports the estimates.²¹ Acquisitions that are followed by further acquisitions in the same CPC group are associated with successively higher patenting: 0.32 after the first and a further 0.17 after the second, both significant at the one percent level, so a group with three or more acquisitions sits about 0.49 above its comparison group once the second has occurred. Acquisitions with no follow-on run the other way: the increment is -0.16 and

²¹The corresponding table is Table B.6 in Appendix B.5.

Figure 10: Matched event study: heterogeneity by acquisition sequence and existence of follow-on acquisition



Notes: Matched estimates of patenting in a CPC group around successive acquisitions, by the acquisition's order in the group's sequence (vertical axis, first through fourth) and by whether a further acquisition follows: *No follow-on* (blue) marks the group's last acquisition, *With follow-on* (orange) an acquisition with at least one later deal in the same CPC group. Acquisitions in the same calendar year count once. The outcome is the arcsinh of annual patent filings. The indicators are cumulative, so each marker gives the change in the difference between event groups and their matched comparison groups relative to the preceding period: the first marker relative to the years before any acquisition, the k -th relative to the period following the $(k - 1)$ -th. The cumulative difference after the k -th acquisition is the sum of the markers up to that point. The specification includes CPC-group and calendar-year fixed effects, with standard errors clustered by CPC group; caps are 95% confidence intervals. Coefficients are reported in Table B.6.

insignificant when a group's only acquisition is its first, but -0.36 after a second acquisition that turns out to be the last and -0.34 after a third, which leaves a group with exactly two acquisitions back at roughly its comparison group's level.

Overall, the evidence in this section shows that patenting grows in a CPC group before its first startup acquisition, that this growth continues in CPC groups where follow-on acquisitions occur, and that the estimated gap relative to matched comparison groups narrows in CPC groups where the acquisition sequence ends (Figures 9 and 10). These patterns are consistent with, though not uniquely implied by, the stylized model. The co-movement of continued growth in patenting and further deals is consistent with the belief-persistence and selection channel of Proposition 1, under which both patenting and acquisition activity

track a common evolving belief rather than one deal mechanically causing the next. The narrowing gap after the last acquisition is consistent with the crowding logic of Proposition 2, which implies innovation is durably harder relative to a no-acquisition benchmark; the proposition characterizes that counterfactual wedge rather than the exact shape of the estimated coefficient path, so we read the match as suggestive rather than as a direct test of the proposition. More broadly, the pattern is consistent with models in which innovation responds to *expectations* about exit or commercialization opportunities. It also points to a limitation of our retrospective analysis. Even if some acquisitions strongly influence subsequent innovation, the more important link between antitrust policy and innovation may run through its ability to shape firms’ ex-ante expectations. Finding exogenous variation in those expectations will be difficult, however, in a lenient environment where most startup acquisitions are approved.

The analysis so far has focused on the quantity of innovation, measured by patent application counts at the CPC group level, and says nothing about the quality, influence, or diffusion of the patents involved. We now turn to citation-based measures, which offer a complementary perspective on how the technological significance and reach of patenting evolve around acquisitions.

5.2 Patent Citation Analysis

In this section, we shift our focus (and unit of analysis) from the CPC group or technology area to the individual patent. Thus, instead of examining the change in overall innovation trends surrounding an acquisition event, we focus on the diffusion and impact of the acquired technology.

5.2.1 Impact of Acquisitions on Acquired Patents

Our main patent level outcome, cit_{pct} , is a count of forward citations to focal patent p , with primary CPC group c , in year t . Citations have been used extensively as a proxy for the technological significance, economic value, and cumulative innovation.

To examine the impact of acquisitions on the importance of acquired patents, we estimate a set of models by Poisson pseudo-maximum likelihood (PPML). The citation count is a non-negative, heavily skewed outcome that is zero in most patent-years. PPML models its conditional mean multiplicatively, which lets the citation response scale with a patent’s baseline citation rate rather than imposing an additive effect, accommodates the zeros directly, and is consistent whenever the conditional mean is correctly specified.²² The linear

²²We also have robustness checks implementing the specification under alternative functional forms,

components resemble those in [Rysman and Simcoe \(2008\)](#).

Specifically, we model the conditional mean of the citation count as

$$\mathbb{E}[cit_{pct} \mid \cdot] = \exp\left(\tau PostAcquisition_{pt} + \beta Acquired_p + \sum_a \delta_a \mathbb{1}\{age_{pt} = a\} + \lambda_{ct}\right) \quad (4)$$

$$\mathbb{E}[cit_{pct} \mid \cdot] = \exp\left(\tau PostAcquisition_{pt} + \delta_2 age_{pt}^2 + \delta_3 age_{pt}^3 + \gamma_p + \lambda_{ct}\right) \quad (5)$$

$$\mathbb{E}[cit_{pct} \mid \cdot] = \exp\left(\sum_{s=-5}^{10} \tau_s d_{p,t=y+s} + \delta_2 age_{pt}^2 + \delta_3 age_{pt}^3 + \gamma_p + \lambda_{ct}\right) \quad (6)$$

where λ_{ct} are CPC-group $\times$ year fixed effects, γ_p patent fixed effects, β the coefficient on the acquired-patent indicator, δ_a (or δ_2 and δ_3) the coefficients capturing the citation age profile, and $s = -1$ (the year before acquisition) is the omitted reference period in (6). Because the count enters through the exponential mean, τ is a semi-elasticity, so that acquisition multiplies a patent's expected citation rate by $\exp(\tau)$, a proportional change of $\exp(\tau) - 1$ (e.g. $\tau \approx 0.22$ implies roughly a 25% increase). We estimate all models by PPML with standard errors clustered at the patent level.

The first specification of equation (4) is a pooled cross sectional model (CS). It includes an indicator $Acquired_p$ for patents owned by the target firm, the diff-in-diff coefficient $PostAcquisition_{pt}$ that equals one for acquired patents after the acquisition occurs, a full set of patent-age effects (where age is calculated relative to grant year) to capture the typical citation life-cycle, and a set of CPC-group by year fixed effects λ_{ct} that capture aggregate time-trends. In equation (5), we introduce a model with patent fixed effects γ_p (WP), which absorb time-invariant differences in patent quality. Because these patent fixed effects also create a potential age-year-cohort co-linearity problem, however, we switch from using age dummies to a pair of terms, age_{pt}^2 and age_{pt}^3 , that capture the non-linear component of the underlying citation age profile. Finally, equation (6) implements an event-study specification, similar to our CPC-year analysis above, in order to estimate a full set of dynamic treatment effects τ_s .²³

Table 7 presents the results from the baseline citation regressions described in specifications (4) and (5), using other patents in treated CPC groups as the control group.²⁴

namely OLS in levels and an inverse hyperbolic sine transformation of the citation count. For more details, see Appendix B.4.

²³To clarify the structure of our citation-level analysis, Section B.2 in the appendix shows and discusses the distribution of treated patents by event year relative to acquisition. The estimation sample differs slightly from the difference-in-differences specifications: the event study includes acquired patents only in the years from five before to ten after the acquisition.

²⁴In Section B.3 in the appendix, we report robustness checks using alternative control groups, including all patents in G and H, and patents assigned to the acquirer.

Table 7: Diff-in-Diff Estimates of Acquisition Impact on Patent Cites

	App-year		Grant-year		Patent Family		App-year (no self-cites)	
	CS (1)	WP (2)	CS (3)	WP (4)	CS (5)	WP (6)	CS (7)	WP (8)
PostAcquisition	0.203*** (0.041)	0.245*** (0.039)	0.223*** (0.053)	0.303*** (0.042)	0.259*** (0.042)	0.318*** (0.050)	0.281*** (0.044)	0.266*** (0.041)
Acquired	0.660*** (0.033)		0.600*** (0.047)		0.579*** (0.031)		0.576*** (0.036)	
Patent FE		✓		✓		✓		✓
CPC×Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Age	i.age	age^2, age^3	i.age	age^2, age^3	i.age	age^2, age^3	i.age	age^2, age^3
Pre-acq mean	1.203	1.341	0.508	0.576	0.881	1.042	1.008	1.163
Pseudo- R^2	0.184	0.522	0.271	0.579	0.217	0.441	0.191	0.522
Observations	45,892,177	36,639,912	45,818,787	35,573,144	41,213,445	31,710,601	45,891,548	35,672,066

Standard errors clustered by patent in parentheses

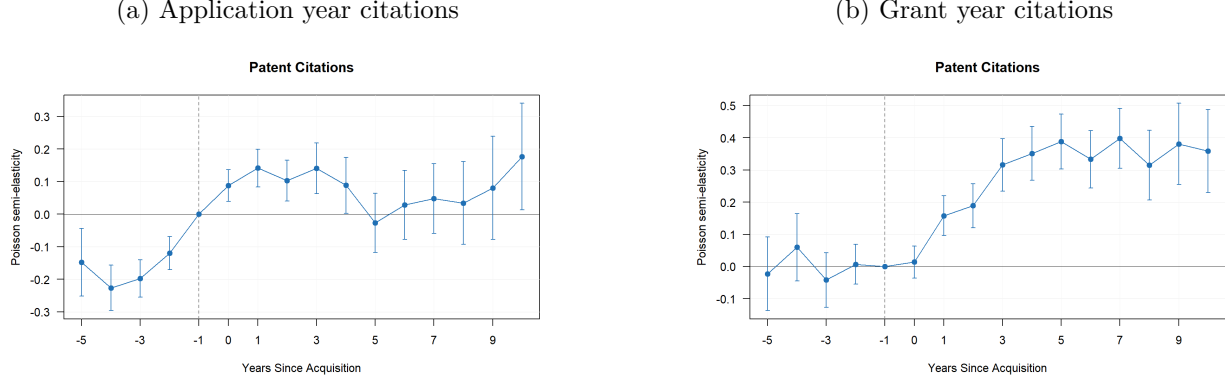
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Standard errors are clustered at the patent level. Column (1) corresponds to a specification with CPC-year fixed effects and a treatment indicator $Acquired_p$, while column (2) replaces the treatment dummy with patent fixed effects to account for time-invariant differences in patent quality and visibility. Both specifications include controls for patent age.

The key variable of interest, $PostAcquisition_{pt}$, is a dummy equal to one in all years after the acquisition of the patent's parent firm. The coefficient on this variable is positive and highly significant in both specifications, indicating that patents filed by acquired firms receive significantly more citations in the years following the acquisition. In column (1), coefficient of 0.20 means that the mean citation rate for acquired patents increases by 23 percent after an acquisition. The coefficient on $Acquired_p$ in column (1) indicates that the patents of acquired startups have a more than 90 percent higher citation rate than other patents in the same CPC classes *before* the acquisition. This suggests that acquired patents are both more important to begin with, and see a large boost in their visibility and significance following an acquisition. In column (2), we add patent fixed effects, and find that the relative increase in citations after an acquisition are, if anything, slightly larger at 28 percent.

Columns (3) through (8) report the same specifications for three alternative citation measures. Columns (3) and (4) date citations to the citing patent's grant year rather than its application year, which matters for the timing of the pre-acquisition trend and which we return to below. Columns (5) and (6) count citations at the level of the DOCDB patent family rather than the individual patent, since the same invention may be patented in several jurisdictions or extended through continuations, producing duplicative citations (Kuhn, Younge and Marco, 2020). Columns (7) and (8) exclude all citations made by the acquiring firm. None changes the picture: in the patent fixed-effects specification the coefficients are

Figure 11: Event study: Application versus Grant Year Cites



Notes: Poisson (`fepois`) event studies of citations received by patents in treated CPC groups around the acquisition. Each panel regresses the citation count on event-time indicators, with patent and $\text{CPC} \times \text{year}$ fixed effects, on the sample of ever-cited patents. Caps are 95% confidence intervals with standard errors clustered by patent.

0.303, 0.318 and 0.266, implying increases of roughly 35, 37 and 30 percent against 28 percent in the baseline. The family-level increase is proportionally slightly larger but, because deduplication lowers the baseline count, smaller in absolute terms.²⁵

The results in Table 7 indicate that acquisitions are followed by a marked increase in the visibility and influence of the acquired firm’s patents. Since the comparison group is composed of other patents in the same CPC groups, these results are not driven by broad shifts in technological fields, but rather reflect changes specific to the individual inventions of the acquired targets. This pattern is consistent with improved dissemination, commercialization, or reuse of the acquired knowledge, and suggests that acquisitions by digital incumbents may facilitate rather than suppress post-merger innovation spillovers.

Figure 11 depicts the estimated τ_s coefficients from the event-study specification (6). The specification includes patent fixed effects, CPC-year fixed effects, and controls for patent age (in quadratic and cubic form), and uses other patents in treated CPC groups as the control group. The event time s indexes years relative to the acquisition of the patent’s parent firm, with $s = 0$ corresponding to the acquisition year.

The difference between the two panels in Figure 11 is driven by an assumption about citation timing. Panel (a) assigns forward citations to the application-year of the citing patent, and panel (b) assigns citations to the year when the citing patent is granted. The

²⁵Figure B.3 in the appendix reports the corresponding event study. The family-level coefficients rise toward the reference year before the acquisition and the increase is concentrated in the first four years, after which the estimates become imprecise. The pooled family-level result is therefore in line with the baseline, while the dynamics carry the same pre-acquisition trend that we discuss below for application-year citations.

convention in the innovation literature is to assign cites to the application-year, based on the idea that this approximates the moment when an inventor utilized her knowledge of the prior art. This inevitably creates some measurement error, however, because citations can be added (by either the applicant or the examiner) during the examination process. If these “in process” citations are caused by an intervening event (e.g., the acquisition of the firm that filed the cited patent), it may be more appropriate to assign citations to the grant-year, as we do in panel (b), to avoid any contamination of the pre-treatment coefficients.

In both panels of Figure 11, we observe a clear and sustained increase in citations to acquired patents following the acquisition. Because the event study is estimated by PPML, the plotted coefficients are semi-elasticities. In the post-acquisition period they settle at roughly 0.4 log points, implying an increase in the citation rate on the order of 50 percent ($\exp(0.4) - 1$). This is broadly consistent with the difference-in-differences results in Table 7, and supports the idea that acquisitions by large digital firms may amplify the diffusion of acquired innovations, rather than curtailing their impact.

While both panels in Figure 11 show an increase in the citation rate around the year of acquisition, the pattern of the dynamic treatment effects differs under the alternative timing assumptions. In panel (a), when forward cites are assigned to the citing patent’s application-year, we observe an increase in the citation rate during the five-year period before the acquisition, followed by a leveling-off two years after the acquisition occurs. In panel (b), when forward cites are assigned to the grant-year, the difference in pre-acquisition trends disappears, and the dynamic treatment effects exhibit a larger and more sustained increase over the post-acquisition years.²⁶

The difference between the two graphs is consistent with the idea that the differential “pre trend” in panel (a) is caused by citations that arrive during the prosecution process (i.e., between the application and grant-year of the citing patent), potentially as a consequence of the acquisition itself. Thus, we interpret the timing of the citation increase in panel (a), along with the disappearance of the pre-trend in Panel (b), as evidence that supports a causal interpretation of our diff-in-diff estimates. Going forward, we follow the convention in the literature and assign citations to the application year of the citing patent.²⁷

²⁶Coefficients are shown for event time -5 to $+10$. We center on this window because the count of observed treated patents drops off steeply outside it (Figure B.2), leaving longer-horizon estimates identified from a small, selected, and compositionally shifting subsample.

²⁷Sticking to the convention of dating forward cites to the application-year of the citing patent makes little difference in later results because we are primarily interested in effect size, as opposed to timing.

Table 8: Patent Citations by Acquisition Sequence

	App-year cites			Grant-year cites		
	(1)	(2)	(3)	(4)	(5)	(6)
PostAcquisition	0.245*** (0.039)	0.221*** (0.046)	0.221*** (0.046)	0.303*** (0.042)	0.280*** (0.047)	0.280*** (0.047)
PostAcq $\times$ First		0.163** (0.073)			0.182* (0.096)	
PostAcq $\times$ First (with follow-on)			0.198** (0.078)			0.353*** (0.105)
PostAcq $\times$ First (no follow-on)			0.069 (0.139)			-0.098 (0.163)
Patent FE	✓	✓	✓	✓	✓	✓
CPC $\times$ Year FE	✓	✓	✓	✓	✓	✓
Age	age^2, age^3	age^2, age^3	age^2, age^3	age^2, age^3	age^2, age^3	age^2, age^3
Pre-acq mean	1.341	1.341	1.341	0.576	0.576	0.576
Pseudo- R^2	0.522	0.522	0.522	0.579	0.579	0.579
Observations	36,639,912	36,639,912	36,639,912	35,573,144	35,573,144	35,573,144

Standard errors clustered by patent in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

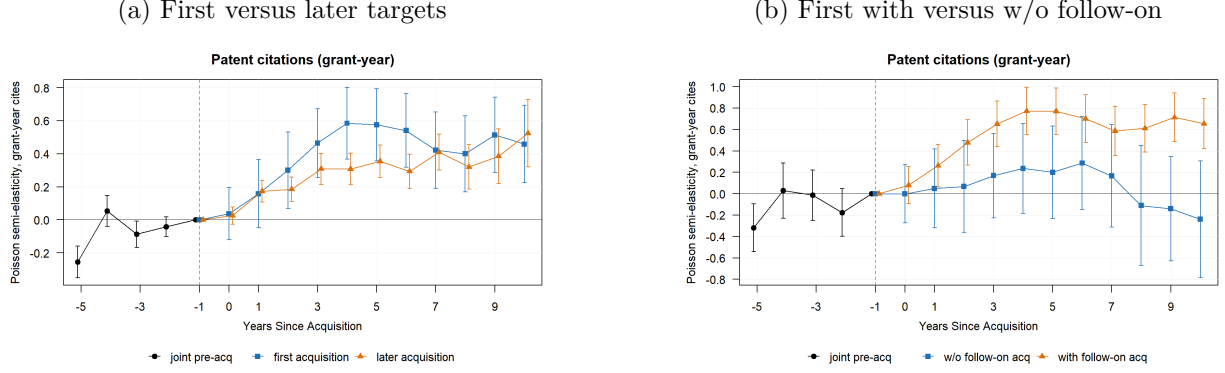
5.2.2 Patent Citation Heterogeneity by Acquisition Sequence

While the average increase in citations suggests that acquisitions can enhance the impact of acquired innovations, this effect may vary depending on the context and sequence of acquisitions. In particular, we explore whether citation dynamics differ between first-acquired targets within a CPC group and those acquired later, and whether the presence of follow-on acquisitions amplifies or moderates these effects. This analysis helps to uncover whether the timing and positioning of an acquisition within a broader strategy shapes the extent to which innovation spillovers materialize.

Table 8 presents estimates from specification (5), where we examine whether the citation effects of acquisitions differ depending on whether the patent belongs to the first acquired target within a CPC group or to a later target. The sample is split accordingly, and we further distinguish between cases where there were follow-on acquisitions and those where there were none. The control group in all regressions remains all other patents in treated CPC groups. The first three columns consider application-year citations, while the last three columns use the number of citations from patents that were granted in a given year.

The interaction term $Post \times First$ captures the effect of an acquisition on patent citations for the first acquired target within a CPC group. $Post \times FirstNoFollowon$ further conditions on there being no subsequent acquisitions in that group, while $Post \times FirstFollowon$ conditions on there being a follow-on acquisition within the same CPC group. Consequently, the coefficient on $PostAcquisition$ in columns (2), (3), (5), and (6) represents the effect for

Figure 12: Patents of first targets within CPC group



Notes: Poisson event studies of grant-year citations for patents in treated CPC groups around the acquisition. Each series is estimated with patent and $\text{CPC} \times \text{year}$ fixed effects. Caps are 95% confidence intervals with standard errors clustered by patent.

patents associated with later acquisition events, that is, targets acquired after an earlier acquisition has already occurred in the CPC group.

The results reveal notable heterogeneity. Patents owned by first-acquired targets in CPC groups that subsequently experience additional acquisitions show a particularly strong post-acquisition increase in citations. This suggests that early movers in acquisition waves may benefit from greater organizational attention or resource integration, which in turn enhances the diffusion of their innovations. In contrast, patents associated with isolated acquisitions (i.e., those not followed by further deals) show smaller or insignificant citation effects, consistent with a more limited post-merger innovation impact. Later-acquired targets also exhibit positive post-acquisition citation effects, though generally smaller in magnitude compared to first-acquired firms in multi-acquisition CPC groups.

These findings locate where the citation increases are concentrated. First acquisitions in areas that go on to see sustained acquisition activity show the strongest citation-based spillovers, the patent-level analogue of the CPC-level pattern.

The left panel of Figure 12 display the estimated τ_s coefficients from the event-study specification (6), separately for patents of first-acquired and later-acquired targets within a CPC group. The right panel presents the coefficients among first-acquired patents only, but separating out whether the first acquisition was followed by another one or not. These figures provide a dynamic view of how the timing of acquisition within a technological domain affects the post-merger trajectory of innovation impact, as measured by forward citations.

The left panel of Figure 12 compares the citation dynamics of first-acquired and later-acquired targets within a CPC group. Patents owned by first-acquired targets show little

movement in citations prior to the acquisition, followed by a pronounced and persistent increase beginning shortly after the acquisition year. For later-acquired targets, the post-acquisition rise is more muted and less sustained. The magnitude of the effect is smaller, and the confidence intervals are wider, especially in later years. These patterns suggest that later acquisitions may receive less organizational focus or integration effort, or may be motivated by defensive rather than developmental aims.

The right panel of Figure 12 restricts attention to first-acquired targets and separates those whose CPC group went on to experience follow-on acquisitions from those that did not. The sharp and sustained rise is concentrated in the former: citations to first-acquired targets in CPC groups with follow-on deals climb steadily over the first four years and remain elevated throughout, mirroring the heterogeneity documented in Table 8. For first acquisitions without follow-on deals, the increase is gradual and never statistically distinguishable from zero, and the point estimates decline at longer horizons.

5.2.3 Sources of Citation Increase and the Role of Retention

Section 5.2.1 showed that the citation increase survives when the acquirer’s own citations are excluded, so it is not simply the acquirer citing its newly acquired patents. This subsection asks which outside firms account for the additional citations, and whether the answer depends on where the target’s inventors end up.

To examine the link between post-acquisition workforce integration and the citation effect, we decompose the citation increase by source into additional citations from (i) the big-tech acquirer, (ii) other firms that hire inventors away from the acquirer’s targets, (iii) other firms that hire those targets’ non-inventor employees, or (iv) firms unrelated to the acquisition.

While the big-tech acquirer definition is straightforward, identifying firms that hire target inventors needs a little care, and similar for other employee destinations. For each target we track the firms at which its former employees took a position in some year after the acquisition in the Revelio work histories. *Inventor destinations* restrict these alumni to confirmed patent inventors so the set collects the post-acquisition employers of the target’s departing inventors. *Other-employer destinations* are the remainder of the pool. These are firms which hired some target alum but to which no confirmed inventor moved and are computed as the broad future-employer set net of the inventor subset.

Both sets are pooled across the targets of a given acquirer rather than constructed separately for each target. Pooling avoids a mechanical problem: a target-specific destination set is empty whenever none of that target’s inventors leave, so its citation count is zero by

construction, and the retention interaction we introduce below would measure whether the channel exists rather than how strongly the departing inventors’ new employers cite the acquired patents. Pooling at the acquirer level keeps the destination sets populated regardless of the retention outcome at any single target. In exchange, the sets describe the acquirer’s post-acquisition talent network as a whole: a firm that hired an inventor from one of the acquirer’s targets counts as an inventor destination for citations to all of that acquirer’s targets. The channel estimates below should therefore be read as citations from firms that hired inventors away from the acquirer’s targets, rather than from the specific firms that hired the focal target’s own inventors.

We estimate each of these regressions by Poisson, so the coefficients are proportional effects, which has two consequences for how the table is read. First, the group citation counts are defined only for acquired patents, while comparison patents keep their total citations. Each column therefore measures a treated patent’s citations from one source against the overall citation trend among comparable patents, and reading the columns as a decomposition requires a proportional version of the common-trends assumption: absent the acquisition, citations from each source would have grown at the same rate as citations overall. The $\text{CPC} \times \text{year}$ fixed effects absorb technology-specific citation trends, which makes this assumption reasonable in our setting. Second, the citation counts of the different groups start from very different baselines, so the proportional effects can neither be compared across columns nor summed. Below each coefficient we therefore report the implied change in citations per patent-year, obtained by applying the proportional effect to the group’s pre-acquisition mean on a common sample of acquired patents. These implied changes, rather than the coefficients themselves, decompose the citation increase.

Table 9 reports the decomposition. In proportional terms the acquirer’s own citation rate rises most steeply, with the coefficient of 1.188 implying that it more than triples. Because the acquirer starts from a low base this is easily read as a small contribution, but in levels it is not: it amounts to 0.12 additional citations per patent-year, close to two-fifths of the total increase. Firms that hired inventors from the acquirer’s targets cite the acquired patents about 32 percent more ($\exp(0.280) - 1$) and firms that hired non-inventor employees about 56 percent more ($\exp(0.447) - 1$), which in levels is 0.04 and 0.18 citations per patent-year respectively; the non-inventor channel is thus not merely larger in proportional terms but roughly four times larger in levels. Firms with no employment link to the acquisition show no increase, with a coefficient close to zero and statistically insignificant. Taken together the employee-destination channels account for around seven-tenths of the increase and the acquirer for most of the remainder, so the additional citations come almost entirely from firms inside the acquisition’s employment orbit, consistent with knowledge spillovers operating

Table 9: Citations by subgroups of companies

	All cites (1)	From acquirer (2)	Inventor destinations (3)	Other employee destinations (4)	From neither (5)
PostAcquisition	0.245*** (0.039)	1.188*** (0.081)	0.280*** (0.063)	0.447*** (0.096)	-0.038 (0.040)
Patent FE	✓	✓	✓	✓	✓
CPC×Year FE	✓	✓	✓	✓	✓
Pre-acq mean	1.203	0.053	0.129	0.314	0.708
Implied Δ cites/yr	0.334	0.121	0.042	0.177	-0.027
Pseudo- R^2	0.522	0.521	0.521	0.522	0.521
Observations	36,639,912	36,590,570	36,596,512	36,617,445	36,623,348

Standard errors clustered by patent in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Citation effect by inventor retention rate at the acquirer

	All cites (1)	From acquirer (2)	Inventor destinations (3)	Other employee destinations (4)	From neither (5)
PostAcquisition	0.352*** (0.042)	1.294*** (0.104)	0.349*** (0.083)	0.380*** (0.079)	0.173*** (0.066)
PostAcquisition × Inventor Retention	0.036 (0.100)	0.619*** (0.234)	-0.459** (0.197)	0.221 (0.179)	-0.087 (0.169)
Patent FE	✓	✓	✓	✓	✓
CPC×Year FE	✓	✓	✓	✓	✓
Pre-acq mean	1.203	0.053	0.129	0.314	0.708
Pseudo- R^2	0.522	0.521	0.521	0.521	0.521
Observations	36,580,396	36,563,525	36,563,503	36,570,183	36,572,760

Standard errors clustered by patent in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

through labor mobility.

The decomposition is consistent with labor mobility, but it does not rule out a simpler explanation: the inventor destinations may simply be the firms technologically closest to the targets, which would cite the acquired patents more regardless of any worker flows. Inventor retention offers a way to distinguish the two. As Section 4.4 shows, retention of target-firm inventors drops sharply in the years following an acquisition, and the extent of the decline varies across deals. If the citation increase reflects the integration of human capital, it should vary with how many inventors remain at the acquirer; if it reflects technological proximity alone, it should not. We therefore interact the post-acquisition indicator with the demeaned share of target inventors still employed at the acquirer one year after the

deal. Retention is measured after the acquisition, is itself shaped by deal characteristics, integration, and outside options, and mechanically governs how many firms populate each destination channel. The estimates therefore document how the composition of the citation increase varies with retention, but they do not identify the mechanism behind it.

Table 10 reports our estimates. At the average retention rate, the post-acquisition coefficient is positive and significant in every column, so the citation increase holds across all channels. For all citations (column 1) the interaction is small and insignificant (0.036), so we cannot detect a relationship between retention and the size of the aggregate increase; this is an absence of evidence rather than evidence that the total is invariant. The channels, however, move in opposite directions. Citations from the acquirer rise with retention (0.619, significant at the 1% level), while citations from inventor destinations fall with retention (-0.459 , significant at 5%). Ten percentage points of additional inventor retention are associated with about 6 percent more citations from the acquirer and about 4.5 percent fewer from inventor destinations.²⁸ The other-employee and unrelated channels do not vary significantly. Where more of the target’s inventors remain at the acquirer, the acquirer’s own citations are higher and citations from inventor destinations lower; where fewer remain, the pattern reverses.

Retention is thus associated with where the additional citations come from rather than, as far as we can detect, with how many there are. When retention is low, more of the target’s inventors leave, and inventor destinations show larger citation increases, consistent with a broader external diffusion of the acquired knowledge and with the Silicon Valley “job hopping” of Fallick, Fleischman and Rebitzer (2006). When retention is high, the inventors remain at the acquirer, whose own citations to the acquired patents are correspondingly higher.

The co-movement is what one would expect if the acquired knowledge travelled with the inventors, but retention is not randomly assigned and may reflect, among other things, how closely the target’s technology matches the acquirer’s. What the estimates document is an association between how many of the target’s inventors stay at the acquirer and whether the additional citations come from the acquirer or from inventor destinations. This suggests that employee departures, often read as a sign of failed integration, need not reduce the diffusion of the acquired technology so much as redirect it beyond the acquirer’s boundaries.

²⁸This second interaction is not robust to dating citations by grant year rather than application year (Table B.8 in the Appendix), a further reason to read the channel split as descriptive.

5.3 Inventor Analysis

We next examine how frequently reliably matched inventors patent in the years surrounding the acquisition. We restrict the sample to a ± 5 -year window around each inventor’s first acquisition and classify each reliable match by employment status one year after the acquisition. An inventor stayed at the acquirer if they are then within the acquirer’s corporate umbrella or still at the target itself. The rest left the acquirer–target pair, and we separate them by when they did so: an inventor left before the acquisition if Revelio does not place them at the target in the year of the acquisition or the year before it, and left after it if they were there at that point but not at the acquirer a year later.²⁹

The timing matters for the comparison. Of the 746 inventors who are not at the acquirer a year later, 463 had already left the target before the deal closed, and only 283 were still there when it did. For those who had already left, the acquisition is not an event in their own career, and they patent less than either other group across the whole window. Pooling them with the inventors who left afterwards therefore pulls that group’s pre-acquisition average down and makes those who stayed appear selected on productivity. We therefore compare the inventors who stayed with those who were still at the target when it was acquired and left afterwards.

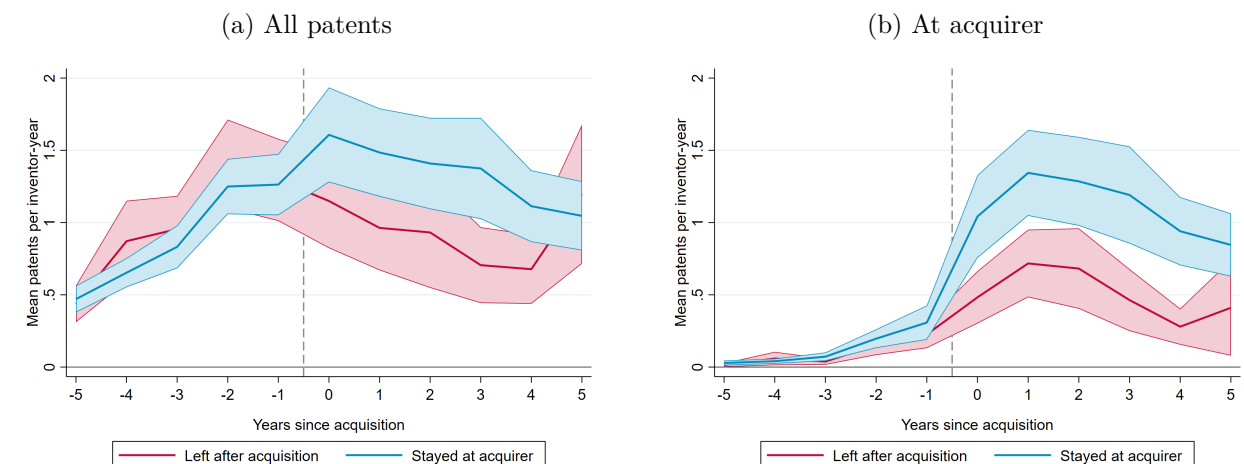
The distinction also matters for the retention rate. Across the full roster of inventors who patented at a target before its acquisition, 49 percent are at the acquirer a year later. Among those still at the target when the deal closed, 72 percent are, ranging from 64 percent at Amazon and Microsoft to 86 percent at Intel.

Prior to acquisition, inventors of acquired start-ups are named on roughly 0.85 patents per year, and only a negligible share of these patents are co-filed with the acquirer.³⁰ Among the two groups that were both at the target when the deal closed, pre-acquisition patenting is very similar: those who left after the acquisition average 0.99 patents per year and those who stayed 0.89, a difference that is insignificant in each of the five years before the acquisition and when the years are pooled ($p = 0.15$). Inventors who had already left the target average 0.70. The groups diverge only after the acquisition. Figure 13 plots mean patenting by year relative to acquisition, separately for the inventors who stayed at the acquirer and those who left after it, with 95% confidence bands. Panel (a) counts all of an inventor’s patents, whereas Panel (b) restricts attention to patents filed under the acquirer’s umbrella.

²⁹Leaving subsumes both new employment at a non-acquirer firm and the absence of any Revelio employment record; the latter accounts for only about ten percent of inventors. Of those we do not place at the target in the year of the acquisition or the year before, most are positively observed at another employer rather than simply absent from the data. Using that two-year window rather than the year before alone avoids reclassifying an inventor on the strength of a single missing Revelio year.

³⁰See Table B.9 in Appendix B.5.

Figure 13: Patent counts around the acquisition, by inventor status.



Notes: For each year relative to a target inventor’s acquisition, the figure plots the mean number of patents on which the inventor is named, separately for inventors who stayed at the acquirer (blue) and those who left after the acquisition (red), with shaded 95% confidence intervals. An inventor counts as having stayed if they are within the acquirer’s umbrella, or still at the target, one year after the acquisition. Both groups were at the target when the acquisition took place; inventors who had already left beforehand are omitted here and reported in the appendix table of group means.

Panel (a) shows that inventors who stayed substantially raise their patenting rate after acquisition, while those who left patent at a lower rate, at least through the fourth year following the event. Panel (b) reveals that the increase for those who stayed is driven almost entirely by patents filed for the acquirer, unsurprisingly, since the acquirer is now their employer. Those who left also show a small rise in acquirer-filed patents, plausibly reflecting persistent collaborations, though it is far less pronounced.

Although these patterns are purely descriptive, they indicate that big-tech acquirers do not merely acquire patents and technology: they also acquire and retain people who, on average, become more productive than before. While we cannot tell whether separations reflect acquirers releasing inventors or inventors leaving voluntarily for better opportunities, the inventors who were at the target when it was acquired were equally productive beforehand whether or not they stayed, so the divergence that follows does not reflect selection on baseline productivity.

6 Conclusion

In this paper we investigated how innovation evolves around acquisitions by major digital incumbents in targeted technological domains. Using a novel dataset that links roughly 1,200

startup acquisitions by eight large technology firms to detailed patent-level and workforce information, we implemented an event-study framework to study changes in both the quantity and quality of innovation before and after acquisition events. By analyzing patenting trends within Cooperative Patent Classification (CPC) groups, and tracking citation patterns to acquired patents, we provided a comprehensive assessment of how technological activity evolves around these deals. To interpret these patterns, we developed a stylized model in which acquisitions and innovation co-move with evolving beliefs about a technology’s commercial viability, generating acquisition waves, a rise in innovation ahead of acquisitions driven by selection, and a durably elevated barrier to standalone innovation relative to what it would have been absent the acquisitions.

Our results offer a new view of the “killer acquisition” narrative. While acquisitions by large digital firms are often viewed with suspicion, we find little evidence that these deals suppress innovation. Instead, patenting activity tends to increase in CPC groups following an acquisition, but only where follow-on acquisitions occur in the same domain: where none follow, patenting flattens or declines, and the gains partly unwind once acquisition activity in an area ends. Matched control group comparisons confirm that this pattern is not driven by differential pre-trends. Citation-based analyses further reinforce these findings. Patents owned by acquired firms receive significantly more citations after the acquisition, and this increase is not solely due to self-citations by the acquirer. Spillovers to external innovators play a meaningful role, suggesting broader knowledge diffusion rather than internal hoarding.

The differences across deals reflect the serial character of acquisition activity that runs through all of our results. Patenting and citations rise most where the acquisition is followed by further deals in the same domain: first acquisitions in areas that go on to see sustained activity show the largest citation increases, and at the CPC level patenting continues to grow only where such follow-on deals occur. Acquirers are in turn far more likely to keep buying in CPC groups where they already held patents, so sustained acquisition activity and an existing technological presence tend to go together. As acquisitions accumulate in an area, the acquirers’ combined share of its patents rises, so these serial strategies also produce a gradual consolidation of patent ownership, the footprint that the model implies makes standalone entry durably harder.

Linking the patent data to workforce histories from Revelio Labs sheds light on the sources of these citation gains. Employee attrition at acquired firms is substantial, with 31% of target employees leaving within a year of the acquisition. While we cannot detect a relationship between the overall post-acquisition citation increase and how many of the target’s inventors are retained, what changes with retention is the *source* of the additional citations. When inventors stay, the acquirer accounts for more of them, and when they leave,

the firms that hire them do. This points to knowledge spillovers that travel with people, diffusing the acquired technology beyond the acquirer’s boundaries through labor mobility.

From a policy perspective, the evidence is more consistent with complementarity than with suppression: innovation does not decline after these deals, and the acquired technologies diffuse more widely. This does not rule out killer acquisitions in specific cases, since shelving is not observable in patent data. Our results suggest, however, that the relevant unit of assessment is the pattern of acquisitions rather than the individual deal: the same acquirers return to the same technology areas, the innovation gains are concentrated in those sustained sequences, and each deal deepens the acquirer’s footprint in the area. This perspective aligns with the attention that the 2023 U.S. Merger Guidelines devote to serial acquisitions. For merger review in digital markets, blanket skepticism toward individual deals may overlook innovation-enhancing integration, while purely deal-by-deal review may miss the cumulative consolidation that only a sequence of acquisitions reveals.

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A Theoretical Model: Setup, Derivations, and Proofs

This appendix provides the full details of the model introduced in Section 2, including the timing and information structure, the derivation of the three acquisition regimes, and the proofs of Propositions 1–3.

A.1 Full Setup

Players and state. Consider a single technology space. Time is discrete, $t = 1, 2, \dots$. There is one incumbent platform firm and one potential entrant per period (short-lived, meaning present for one period only). The space has an unknown *commercial viability* state $\theta \in \{H, L\}$, with prior $\mu_0 \equiv \Pr(\theta = H) \in (0, 1)$. A project with technical quality x has commercial value x if $\theta = H$ and zero if $\theta = L$. Quality does not depend on θ . The state determines whether innovations are valuable, not whether they are technically feasible.

Timing. Let N_t denote the cumulative number of acquisitions before period t . Each period proceeds in four steps.

1. *Signal.* A public signal $s_t \in \{h, \ell\}$ about θ is drawn, with $\Pr(s_t = h \mid \theta = H) = q_H > q_L = \Pr(s_t = h \mid \theta = L)$, independent across periods. All participants observe s_t and update beliefs to μ_t via Bayes' rule.
2. *Quality draw.* The entrant draws project quality $x_t \geq 0$ from F (full support on $\mathbb{R}_+$, independent of θ).
3. *Entry.* The entrant observes x_t and decides whether to develop at cost $c_E > 0$.
4. *Observation and acquisition.* Whether development occurred, and if so the quality, become public. If the entrant developed, the incumbent may acquire the project.

Beliefs. The belief μ_t incorporates all signals $s_1, \dots, s_t$. Given the prior belief μ_{t-1} (before the period- t signal), Bayes' rule gives the update

$$\mu_t = \begin{cases} \frac{q_H \mu_{t-1}}{q_H \mu_{t-1} + q_L (1 - \mu_{t-1})} > \mu_{t-1} & \text{if } s_t = h, \\ \frac{(1 - q_H) \mu_{t-1}}{(1 - q_H) \mu_{t-1} + (1 - q_L)(1 - \mu_{t-1})} < \mu_{t-1} & \text{if } s_t = \ell. \end{cases} \quad (7)$$

We write $\mu_t^h(\mu_{t-1})$ and $\mu_t^\ell(\mu_{t-1})$ for the two possible realizations. Since quality is independent of θ , the development and acquisition decisions carry no additional information about θ

beyond what is in μ_t .

Project values. The value of a quality- x project depends on who commercializes it and how many acquisitions the incumbent has already made in the space. Commercialization yields $\mu_t \cdot \kappa(N_t) \cdot x$, where $\kappa(N_t)$ is the *commercialization advantage* with $\kappa(0) > 1$, reflecting complementary assets. An incumbent that acquires and commercializes additionally pays an irreversible development cost $c_I > 0$. If the entrant develops and operates independently, it receives a *standalone value* $\mu_t \cdot \gamma(N_t) \cdot x$, where $\gamma(N_t)$ captures *competitive crowding*, with $\gamma(0) = 1$ and γ non-increasing in N . Standalone entry also imposes a *competitive externality* $\mu_t \cdot \psi(N_t) \cdot x$ on the incumbent, where $\psi(N_t) \geq 0$. A standalone competitor erodes the value of the incumbent's existing portfolio.³¹ Throughout we maintain $\kappa(N) > 0$ and $\gamma(N) > 0$ for all $N \geq 0$; this keeps the entry threshold (1) and commercialization threshold (10) finite and strictly positive at every N , and rules out the regime boundaries in the next paragraph from becoming degenerate.

Payoffs. If the entrant does not develop, all payoffs are zero. If the entrant develops, the incumbent makes a take-it-or-leave-it acquisition offer at price

$$P_t = \mu_t \gamma(N_t) x_t, \quad (8)$$

exactly matching the entrant's standalone payoff.³² The entrant accepts when indifferent, and the incumbent chooses whether to offer at all, and, conditional on acquiring, whether to *commercialize* the project (paying c_I and earning $\mu_t \kappa(N_t) x_t$) or *shelve* it (earning nothing beyond the price paid). The incumbent's net surplus from acquisition, relative to no deal (entrant standalone, incumbent absorbs the damage $\mu_t \psi(N_t) x_t$), is

$$S_t = \max(S_t^C, S_t^S, 0), \quad (9)$$

with $S_t^C \equiv \mu_t x_t [\kappa(N_t) + \psi(N_t) - \gamma(N_t)] - c_I$ (commercialization surplus) and $S_t^S \equiv \mu_t x_t [\psi(N_t) - \gamma(N_t)]$ (shelving surplus). The incumbent acquires whenever $S_t > 0$ and chooses commercialization iff $S_t^C \geq S_t^S$, i.e., iff $x_t \geq c_I / (\mu_t \kappa(N_t))$.

Three regimes. The parameter space $(\kappa(N_t), \psi(N_t), \gamma(N_t))$ partitions into three cases.

³¹One can also let γ and ψ depend on the number of standalone products in the market.

³²Equivalently, the incumbent has all bargaining power. Giving the entrant a share of the surplus would add an "entry-for-buyout" channel that reinforces the model's qualitative predictions without adding new ones.

- *Pre-emption regime*, where $\psi(N_t) > \gamma(N_t)$. In this case $S_t^S > 0$ for every $x_t > 0$, so the incumbent acquires *every* developed project, commercializing those with $x_t \geq c_I/(\mu_t \kappa(N_t))$ and shelving the rest. The shelved deals constitute *killer acquisitions*.
- *Commercialization regime*, where $\psi(N_t) \leq \gamma(N_t)$ and $\kappa(N_t) + \psi(N_t) > \gamma(N_t)$. Here $S_t^S \leq 0$ but $S_t^C > 0$ for x_t large enough, so the incumbent acquires only projects with $x_t \geq c_I/(\mu_t[\kappa(N_t) + \psi(N_t) - \gamma(N_t)])$ and commercializes all of them.
- *No-acquisition regime*, where $\kappa(N_t) + \psi(N_t) \leq \gamma(N_t)$. Here $\max(S_t^C, S_t^S) \leq 0$ for every $x_t > 0$, so the entrant, if it develops, keeps the standalone value.

Entry condition. Since $P_t = \mu_t \gamma(N_t) x_t$ equals the entrant's standalone value, the entrant's development payoff is $\mu_t \gamma(N_t) x_t - c_E$ whether or not the project is acquired. The entrant therefore develops iff $x_t \geq \underline{x}(\mu_t, N_t) \equiv c_E/[\mu_t \gamma(N_t)]$, the threshold reported as (1) in the main text. The threshold is strictly decreasing in μ_t and weakly increasing in N_t through the crowding parameter γ .

Which developed projects get acquired still depends on the regime. In the pre-emption regime, the incumbent acquires every developed project ($S_t^S > 0$ for every $x_t > 0$). In the commercialization regime, the incumbent acquires iff $S_t^C \geq 0$, i.e., iff

$$x_t \geq x^*(\mu_t, N_t) \equiv \frac{c_I}{\mu_t[\kappa(N_t) + \psi(N_t) - \gamma(N_t)]}. \quad (10)$$

Developed projects with $x_t \in [\underline{x}, x^*)$ are therefore developed but not acquired, and produce standalone entry. In the no-acquisition regime, no project is acquired at any quality.

It is convenient to collect these cases in a single *acquisition threshold*

$$x^\dagger(\mu_t, N_t) \equiv \begin{cases} \underline{x}(\mu_t, N_t) & \text{in the pre-emption regime,} \\ x^*(\mu_t, N_t) & \text{in the commercialization regime,} \\ +\infty & \text{in the no-acquisition regime,} \end{cases} \quad (11)$$

so that $a_t = 1$ iff $x_t \geq x^\dagger(\mu_t, N_t)$. Because the regime is determined by N_t alone and does not depend on μ_t , $x^\dagger(\cdot, N_t)$ is a single well-defined function of μ_t for each fixed N_t : it coincides with $\underline{x}$ or x^* , both strictly decreasing in μ_t wherever finite, and is constant at $+\infty$ in the no-acquisition regime. Hence $x^\dagger(\mu_t, N_t)$ is weakly decreasing in μ_t for every N_t , strictly so whenever an acquisition regime is active.

Assumption 1. $\kappa(N) + \psi(N) - \gamma(N)$ is decreasing in N .

Assumption 1 requires only that the combined surplus $\kappa + \psi - \gamma$ decline in N , so each additional acquisition shrinks the surplus available for the next. It does not pin down which of the three primitives does the work. A natural mechanism is the Arrow replacement effect operating on κ , where each additional acquisition in the same technology space cannibalizes rents from earlier ones and shrinks the marginal commercialization value to the incumbent. The assumption is equally satisfied without any decline in κ , provided ψ erodes fast enough as the incumbent's portfolio expands. We do not require the combined surplus to eventually vanish. Whether and when it does depends on the shape of κ , γ , and ψ , and Propositions 2 and 3 therefore treat the termination of the wave and the emergence of a pre-emption phase as conditional events rather than assumed ones.

A.2 Derivations and Proofs

By (11), the acquisition threshold $x^\dagger(\mu_t, N_t)$ is weakly decreasing in μ_t for fixed N_t , strictly so whenever an acquisition regime is active, so higher beliefs (weakly) encourage acquisition. Since $\mu_t^h > \mu_{t-1} > \mu_t^\ell$, acquisition is (weakly) more likely after h -signals. Averaging over the signal realization, acquisition is (weakly) more likely under $\theta = H$. Specifically,

$$\Pr(a_t=1 \mid \theta, \mu_{t-1}, N_t) = q_\theta(1-F(x^\dagger(\mu_t^h, N_t))) + (1-q_\theta)(1-F(x^\dagger(\mu_t^\ell, N_t))), \quad (12)$$

which is (weakly) larger under H since $q_H > q_L$ and $1-F(x^\dagger(\mu_t^h, N_t)) \geq 1-F(x^\dagger(\mu_t^\ell, N_t))$, strictly so whenever an acquisition regime is active at N_t . Moreover, $\mathbb{E}[\mu_t \mid \theta=H, \mu_{t-1}] - \mathbb{E}[\mu_t \mid \theta=L, \mu_{t-1}] = (q_H - q_L)(\mu_t^h - \mu_t^\ell) > 0$, so beliefs drift toward the truth. Acquisitions thus occur disproportionately in $\theta = H$ spaces, creating the appearance of a pre-trend.

Proof of Proposition 1. Part (i). Higher μ_{t-1} implies higher μ_t^h and μ_t^ℓ (Bayes' rule (7) is monotone), which weakly lowers the acquisition threshold $x^\dagger$ in (11) (strictly so whenever an acquisition regime is active at N_t), so (12) weakly rises, strictly so in that case.

Part (ii). By (12) and $q_H > q_L$, $\Pr(a_\tau = 1 \mid \theta = H) \geq \Pr(a_\tau = 1 \mid \theta = L)$, strictly so whenever an acquisition regime is active at N_τ ; we maintain this throughout. Bayes' rule applied to the event $a_\tau = 1$ gives $\Pr(\theta = H \mid a_\tau = 1) > \Pr(\theta = H)$.

For the belief path, fix $t < \tau$. By Part (i), $\pi(\mu_{\tau-1}) \equiv \Pr(a_\tau = 1 \mid \mu_{\tau-1})$ (integrating out s_τ and x_τ) is a nondecreasing function of $\mu_{\tau-1}$. Conditioning on $a_\tau = 1$ reweights the law of $\mu_{\tau-1}$ by $\pi(\mu_{\tau-1})/\Pr(a_\tau = 1)$, a nondecreasing weight, which shifts $\mu_{\tau-1}$ up in the sense of first-order stochastic dominance: for any nondecreasing ϕ , $\text{Cov}(\phi(\mu_{\tau-1}), \pi(\mu_{\tau-1})) \geq 0$, which rearranges (dividing by $\mathbb{E}[\pi(\mu_{\tau-1})] = \Pr(a_\tau = 1)$) to $\mathbb{E}[\phi(\mu_{\tau-1}) \mid a_\tau = 1] \geq \mathbb{E}[\phi(\mu_{\tau-1})]$.

We extend this to every $t < \tau - 1$ by backward induction on the belief chain. The

one-period transition from μ_t to μ_{t+1} is stochastically monotone in μ_t : both realizations $\mu_{t+1}^h(\mu_t)$ and $\mu_{t+1}^\ell(\mu_t)$ are increasing in μ_t by (7), and the probability of the higher realization, $\Pr(s_{t+1} = h \mid \mu_t) = \mu_t q_H + (1 - \mu_t) q_L$, is itself increasing in μ_t since $q_H > q_L$. A stochastically monotone Markov kernel propagates first-order stochastic dominance backward: if the reweighted law of μ_{t+1} dominates its unconditional law, the law of μ_t implied by the same reweighting (through the kernel connecting μ_t to μ_{t+1}) dominates its unconditional law as well (cf. [Milgrom, 1981](#)). Inducting backward from $t = \tau - 1$ gives that $\mu_t \mid a_\tau = 1$ first-order stochastically dominates μ_t unconditionally for every $t < \tau$. Since the acquisition probability at t , $1 - F(x^\dagger(\mu_t, N_t))$, is itself a nondecreasing function of μ_t (same monotonicity as Part (i)), FOSD dominance gives $\mathbb{E}[1 - F(x^\dagger(\mu_t, N_t)) \mid a_\tau = 1] > \mathbb{E}[1 - F(x^\dagger(\mu_t, N_t))]$ for every $t < \tau$: the expected acquisition probability at every $t < \tau$, conditional on an acquisition at τ , exceeds its unconditional counterpart. $\square$

Under H , beliefs drift upward on average and the acquisition probability in (12) tends to rise with them, sustaining a run of deals even as each completed acquisition raises N_t and, under Assumption 1, weakly lowers the surplus available for the next one. Under L , ℓ -signals are more frequent, beliefs decline on average, and the wave is more likely to stall. Whether the combined surplus $\kappa + \psi - \gamma$ actually reaches the no-acquisition boundary is a separate question determined by the shape of the primitives; Assumption 1 alone does not guarantee it, and acquisitions cease only if and when it does. A wave of acquisitions, however long, leaves behind an entrenched incumbent: because γ is weakly decreasing in N , each acquisition weakly raises the acquisition threshold, an effect that persists after the wave ends, since N freezes at N_{T^*} and so does the eroded γ .

Proof of Proposition 2. Immediate from (1) and the monotonicity of γ , namely $\gamma(N_t) \leq \gamma(0) = 1$, with strict inequality whenever γ has declined. If acquisitions cease at T^* (i.e., $\kappa(N_{T^*}) + \psi(N_{T^*}) \leq \gamma(N_{T^*})$), Assumption 1 together with N 's non-decreasing path implies $\kappa(N) + \psi(N) \leq \gamma(N)$ for all $N \geq N_{T^*}$, so no further acquisitions occur, N freezes at N_{T^*} , and $\gamma(N_t) = \gamma(N_{T^*})$ for all $t \geq T^*$. $\square$

The incumbent's moat is larger after longer waves, since more acquisitions weakly lower γ , which weakly widens the entry-threshold wedge relative to the no-acquisition counterfactual at a common belief. This is a counterfactual comparison, not a claim about the realized time path: because μ_t continues to evolve after T^* , higher subsequent beliefs can offset the lower γ , so realized post-wave innovation need not fall below its own pre-wave level even though it remains durably lower than it would have been absent the acquisitions.

Proof of Proposition 3. (i) $\psi(N_t) > \gamma(N_t)$ implies $S_t^S = \mu_t x_t [\psi - \gamma] > 0$ for all $x_t > 0$, so every developed project has strictly positive acquisition surplus. (ii) The incumbent commercializes iff $S_t^C \geq S_t^S$, i.e., iff $x_t \geq c_I/(\mu_t \kappa(N_t))$, and shelves otherwise. (iii) As $\kappa(N_t)$ falls, the commercialization threshold $\bar{x}(\mu_t, N_t) = c_I/(\mu_t \kappa(N_t))$ rises directly from this expression, so a weakly larger range of developed-project qualities is shelved. This is a statement about the threshold $\bar{x}(\mu_t, N_t)$ alone; it does not by itself sign the change in the conditional share of acquired projects that are shelved, since the conditioning set is bounded below by the entry threshold $\underline{x}(\mu_t, N_t) = c_E/[\mu_t \gamma(N_t)]$, which is also increasing in N_t whenever γ declines. $\square$

Discussion. The model produces two distinct senses in which technology spaces dominated by the incumbent are hostile to standalone innovation. The first is a passive byproduct of the wave. By Proposition 2, accumulated competitive crowding ($\gamma(N_t) < 1$) raises the independent-development threshold both during and after the wave. The second is direct. In the pre-emption phase (Proposition 3), the incumbent acquires every developed project and shelves the lower-quality ones outright as killer acquisitions. Technology spaces where the incumbent has been most active are therefore the least hospitable to standalone innovation, through two mechanically distinct channels.

B Empirical Robustness Checks

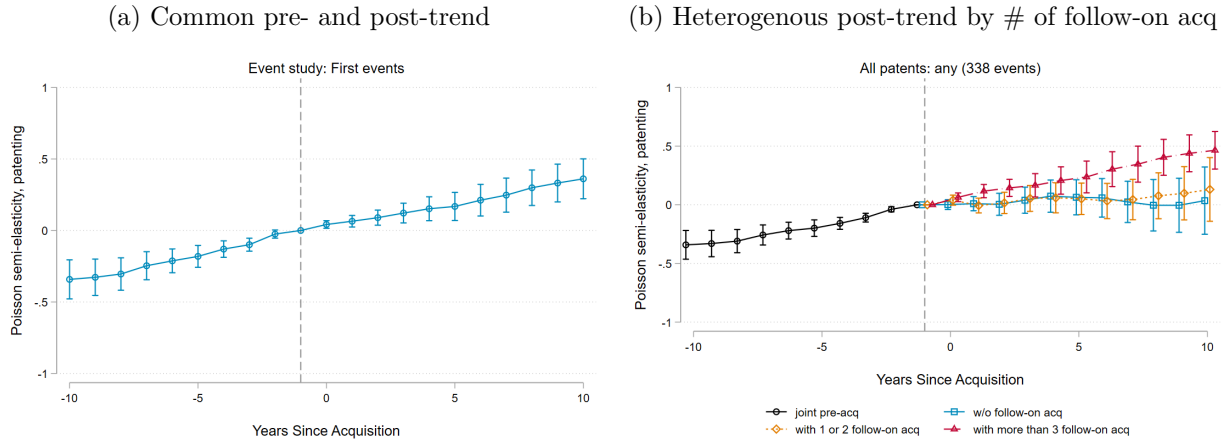
B.1 Poisson for CPC-Level Analysis

In contrast to the citation-level analysis below, where we estimate Poisson models, we use the more conventional inverse hyperbolic sine (asinh) transformation for the CPC-level analysis. This is more defensible here than at the citation level. The large majority of CPC groups record at least one patent application in any given year, and among CPC groups that experience an acquisition only about 1% of CPC-year observations are zero. Because so few observations lie at the zero/positive boundary, treatment can move only a negligible share of the sample across the extensive margin, so the sensitivity of the acquisition coefficient to the units of the outcome—the concern raised by [Chen and Roth \(2024\)](#)—is unlikely to be material here.

As a robustness check, we re-estimate the CPC-level event study with a Poisson pseudo-maximum-likelihood (PPML) estimator on the raw patent count. The results are barely distinguishable from the asinh specification. Figure B.1 reports the Poisson event study. It plots the joint pre-acquisition trend together with the post-acquisition path, split by the

number of follow-on acquisitions.

Figure B.1: Poisson (PPML) event study: first acquisitions within the CPC group



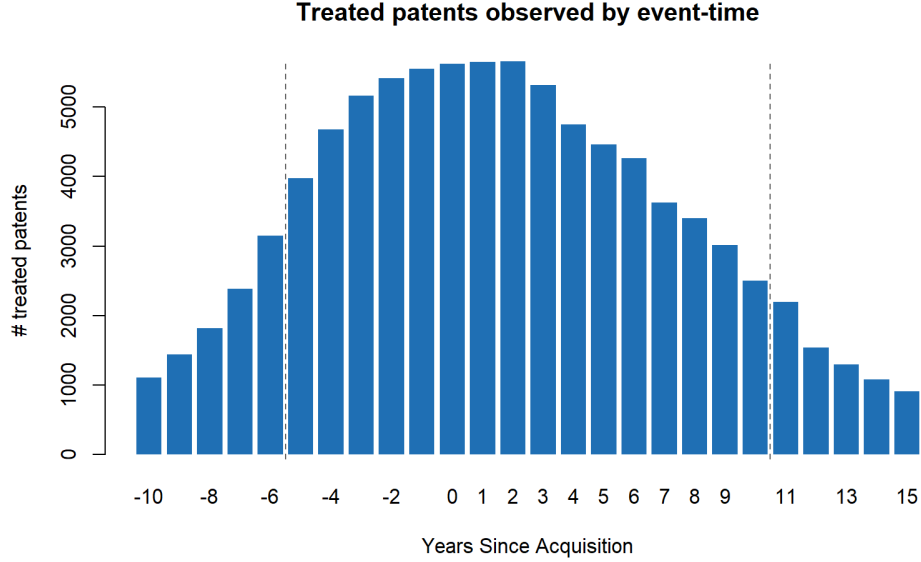
B.2 Citation Analysis Structure

To better understand the structure of the citation analysis, Figure B.2 plots the number of treated patents by event time (i.e., by the number of years before or after the acquisition of their parent firm). This figure provides insight into the distribution of observation counts across event years in the citation-level regressions. The horizontal axis indicates the number of years relative to the acquisition event (with zero denoting the year of acquisition), while the vertical axis shows the total number of patent-year observations in each relative year.

The distribution is concentrated in the period spanning roughly five years before to five years after acquisition. This reflects the construction of the panel: each patent contributes observations from five years before to twenty years after grant, truncated at 2022, so patents acquired late in the sample contribute fewer post-acquisition years and coverage thins at longer horizons on both sides.

The event-study specifications use a window of five years before to ten years after the acquisition, marked by the dashed lines in Figure B.2. The window follows from the coverage the panel provides rather than being set a priori. Each patent enters the panel from five years before to twenty years after its grant, truncated at 2022, so event-time coverage falls away on both sides: patents acquired soon after grant contribute few pre-acquisition years, and patents acquired late in the sample contribute few post-acquisition years. The figure plots the number of treated patents observed at each event year over a wider range, and the counts collapse outside the bracketed window.

Figure B.2: Number of treated patents by year relative to acquisition



B.3 Different Control Groups for Patent Citations

Table B.1 collects regression results using different control groups for specification (5). All estimates are similar in magnitude and significance to our main results. Column (1) presents our main specification, in which acquired patents are compared with other patents in the same CPC group and year. Column (2) runs the same regression on the full sample of non-acquired patents rather than only those in treated CPC groups. The coefficient barely changes, which is what one would expect: with CPC-group-by-year fixed effects the comparison is still drawn within CPC groups, so widening the sample adds observations without altering the counterfactual.

Column (3) instead uses the acquiring firm's own patents as the control group, adding acquirer-by-year fixed effects so that each acquired patent is compared with the same acquirer's patents in the same year and CPC group. This tests whether the post-acquisition increase reflects general firm-level dynamics – an overall rise in patent visibility, increased marketing, or changes in citation practices within large firms – or whether the acquired patents respond distinctly; with the acquirer and year held fixed, a firm-wide citation shock cannot drive the estimate. The comparison is available for almost all treated patents, since 94 percent of acquired patent-years share a CPC-group-by-year cell with their own acquirer's patents. The coefficient of 0.230 is close to the baseline, so the citation gains are not merely part of a firm-wide trend: compared directly with the acquirer's own innovation output, acquired patents receive a larger citation boost.

Table B.1: Different Control Groups

	Treated CPC (1)	All CPC (2)	Acquirer (3)
PostAcquisition	0.245*** (0.039)	0.236*** (0.040)	0.230*** (0.043)
Patent ID FE	✓	✓	✓
CPC×Year FE	✓	✓	✓
Acquirer×Year FE			✓
Pre-acq mean	1.341	1.339	1.339
Pseudo- R^2	0.522	0.511	0.562
Observations	36,639,912	50,093,552	2,065,145

Standard errors clustered by patent in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B.4 Different Regression Models for Patent Citations

The number of citations a patent receives in a given year is a non-negative integer count that is highly right-skewed and contains a large mass of zeros, with most patents receiving no citations at all in any given year. The estimated effect of acquisition therefore depends materially on how the citation outcome enters the regression, and we consider three functional forms. Table B.2 reports the headline within-patent specification under each.

Linear model (OLS). The most transparent option regresses the citation *count* on the post-acquisition indicator and fixed effects by ordinary least squares, so the coefficient is an effect in citations per year. This has two drawbacks for our setting. First, it imposes an additive treatment effect and baseline trend, hence assumes that changes in citations are not related to the baseline number of citations. For a heavily skewed count this is economically implausible and it leaves the estimate sensitive to the small number of very highly cited patents in the right tail. Second, the linear model does not respect the non-negativity of the outcome.

Inverse hyperbolic sine. A common remedy for skewed, zero-valued outcomes is to transform the dependent variable with the inverse hyperbolic sine, $\text{asinh}(y) = \log(y + \sqrt{y^2 + 1})$, which behaves like $\log y$ for large y but is defined at $y = 0$, and to read the coefficient as an approximate percentage effect. Chen and Roth (2024) show, however, that this interpretation is not innocuous. When treatment shifts the extensive margin (whether a patent is cited at all) the coefficient on $\text{asinh}(y)$ is sensitive to the units in which y is measured, so rescaling the outcome changes the estimated effect and it cannot be interpreted as a unit-free elasticity. Our citation panel is exactly the case the critique targets, since zeros are pervasive and acquisition plausibly moves patents on and off the extensive margin. We therefore report the asinh specification only as a robustness check and do not interpret its magnitude.

Poisson (PPML), our main specification. We adopt the fixed-effects Poisson pseudo-maximum-likelihood estimator, which models the conditional mean multiplicatively, $\mathbb{E}[\text{cites}_{it} | \cdot] = \exp(\beta \text{Post}_{it} + \gamma_i + \delta_{c(i)t})$ with patent fixed effects γ_i and CPC $\times$ year fixed effects $\delta_{c(i)t}$. This choice addresses both problems above. The estimator operates on the untransformed count, so the zeros enter directly and require no ad hoc transformation. The coefficient β is a semi-elasticity. It measures the *proportional* change in expected citations ($e^\beta - 1$) and is invariant to the units of the outcome, so it is interpretable in a way the asinh coefficient is not. And by the standard pseudo-maximum-likelihood argument, PPML is consistent under the single assumption that the conditional mean is correctly specified, and it does

not require citations to be Poisson-distributed or equidispersed. We pair the estimator with patent-clustered standard errors. The multiplicative form is also the economically natural one for a skewed count, allowing the absolute citation response to scale with a patent’s baseline citation propensity.

One practical consequence is that, with patent fixed effects, the Poisson estimator drops patents that are never cited, which contribute no within-patent variation. Its sample is therefore the set of ever-cited patents, whereas the OLS and asinh columns retain the full set of treated-CPC patents (the zeros are valid observations for them). Table B.2 reports each column’s sample, pre-acquisition mean, and number of observations accordingly. Across the three estimators the estimated post-acquisition effect is consistent in sign and significance, so our finding is not an artifact of functional form, and we report the Poisson specification throughout for the reasons above.

Table B.2: Functional-form robustness: Poisson, OLS, and arcsinh

	Poisson (1) PPML	Citations (2) OLS	asinh(Citations) (3) asinh
PostAcquisition	0.245*** (0.039)	1.024*** (0.082)	0.195*** (0.009)
Patent ID FE	✓	✓	✓
CPC×Year FE	✓	✓	✓
Sample	ever-cited	all	all
Pre-acq mean	1.341	1.203	1.203
Pseudo- R^2	0.522	—	—
Adj. R^2	—	0.421	0.471
Observations	36,639,912	45,904,452	45,904,452

Standard errors clustered by patent in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B.5 Additional Figures and Tables

This section provides supplementary figures and tables that complement and extend the main results presented in the paper.

Table B.3: Excluded deals

Target	Acquirer	Year	Pre-acquisition patents
Intel smartphone modem	Apple	2019	42,876
Hewlett Packard (Palm)	Qualcomm	2014	32,459
Texas Instrument - Cable Modem	Intel	2020	24,733
Motorola Mobility	Google	2012	21,764
Nokia - Devices & Services Business	Microsoft	2013	14,623
AMD	Qualcomm	2009	9,910
Infineon Technologies (Wireless Solutions Business)	Intel	2010	8,364
LSI - Axxia Networking Business	Intel	2014	6,084
Altera	Intel	2015	3,916
Eaton Corporation (FleetAdvisor)	Qualcomm	2001	3,487
HTC (Pixel Phone Division)	Google	2018	1,782
CSR	Qualcomm	2015	1,277
Dialog Semiconductor	Apple	2019	837
Scientific-Atlanta, Inc.	Cisco	2005	671
McAfee	Intel	2010	593
Qlogic - InfiniBand	Intel	2012	266
Lantiq	Intel	2015	219
Metaswitch Networks	Microsoft	2020	164
Tandberg	Cisco	2010	90

Table B.4: Dealsize, patents and target size

	(1) Deal size (\$M)
Target has patents	461.63** (226.36)
# employees	0.54*** (0.00)
Target has patents $\times$ # employees	1.20*** (0.08)
Constant	266.83*** (90.84)
Observations	190

Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure B.3: Patent family citations

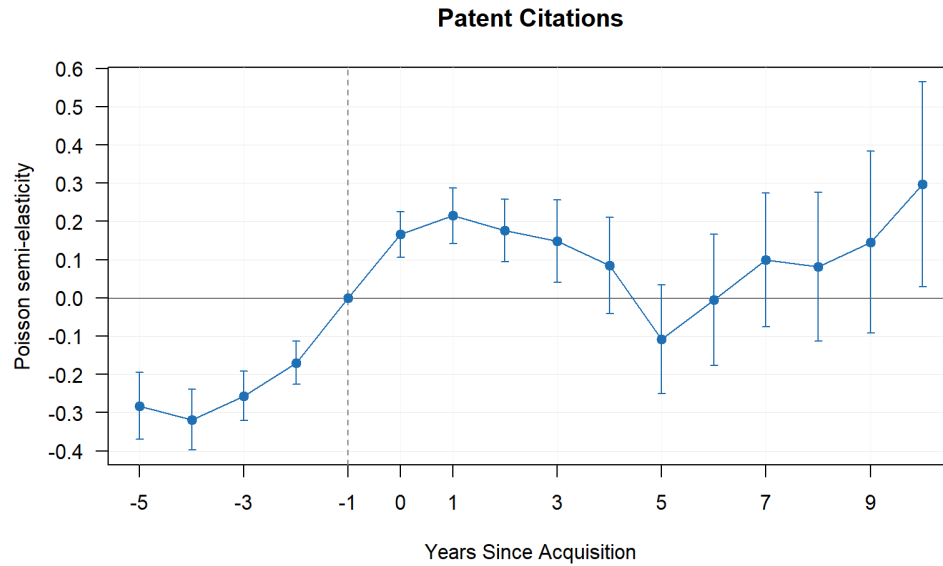


Table B.5: Selection into the Revelio-linked sample

	Matched to Revelio (1)	Not matched (2)	Difference (2) – (1)
<i>Panel A: all post-2005 acquisitions</i>			
Holds pre-acquisition patents	0.48	0.18	-0.30***
Pre-acquisition patents	9.50	2.38	-7.13***
Deal size, USD millions (where disclosed)	977.19	70.23	-906.96***
Acquisition year	2013.29	2011.83	-1.47***
Target outside the US	0.29	0.33	0.04
Acquisitions	692	240	932
<i>Panel B: targets holding pre-acquisition patents</i>			
Pre-acquisition patents	19.81	13.28	-6.53
Deal size, USD millions (where disclosed)	1219.01	96.10	-1122.91***
Acquisition year	2013.42	2011.44	-1.98***
Target outside the US	0.27	0.16	-0.11*
Acquisitions	332	43	375

Difference column reports two-sample t -tests (unequal variances)

Deal size is disclosed for 260 acquisitions in Panel A and 149 in Panel B

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure B.4: Years to next own acquisition within the same CPC group

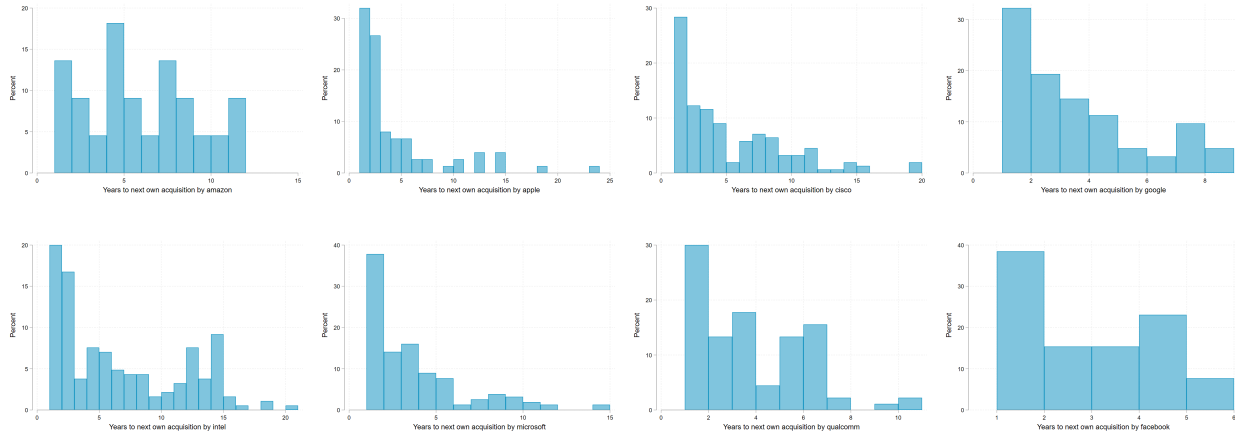


Figure B.5: Years to next acquisition, addressing selection

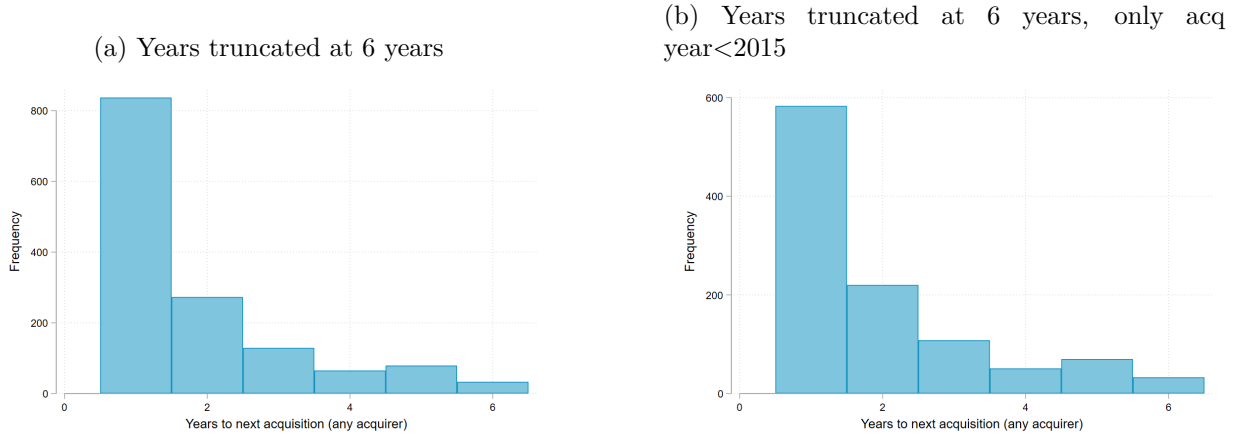


Figure B.6: First acquisition events within CPC group

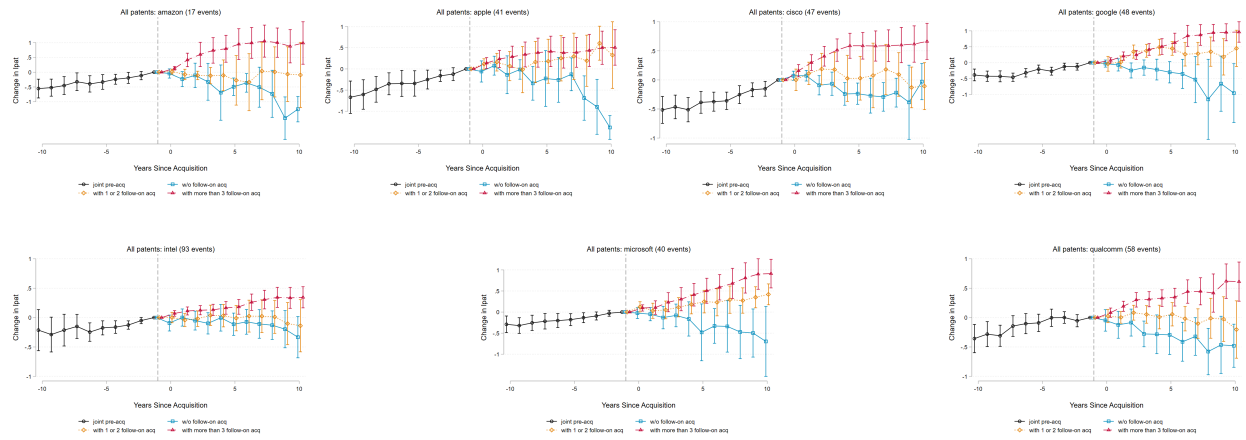


Figure B.7: Matched event studies; All first acquisition events

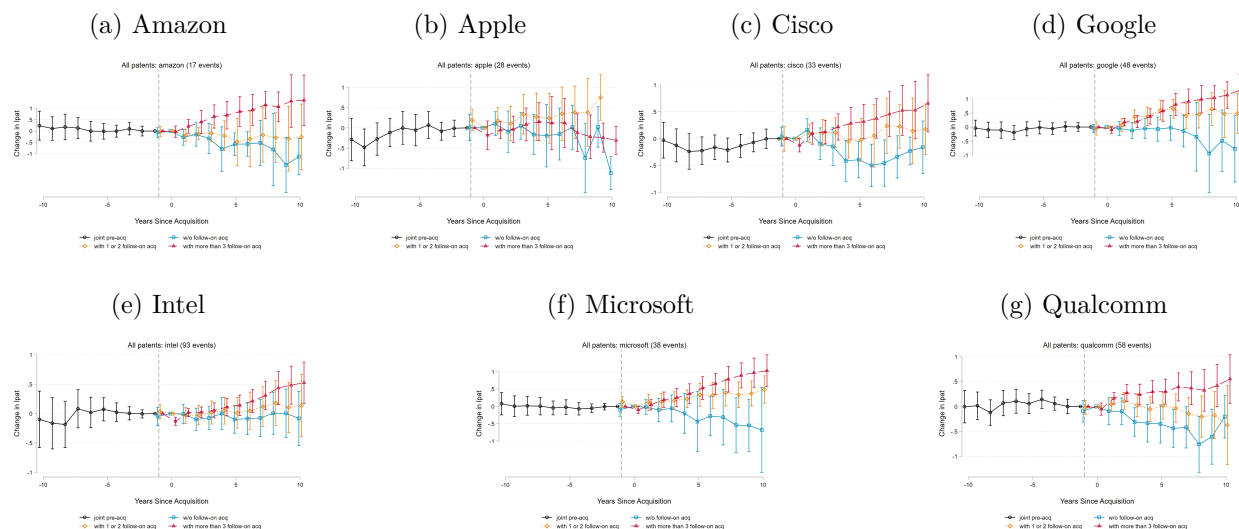


Figure B.8: Retention rates by acquirer

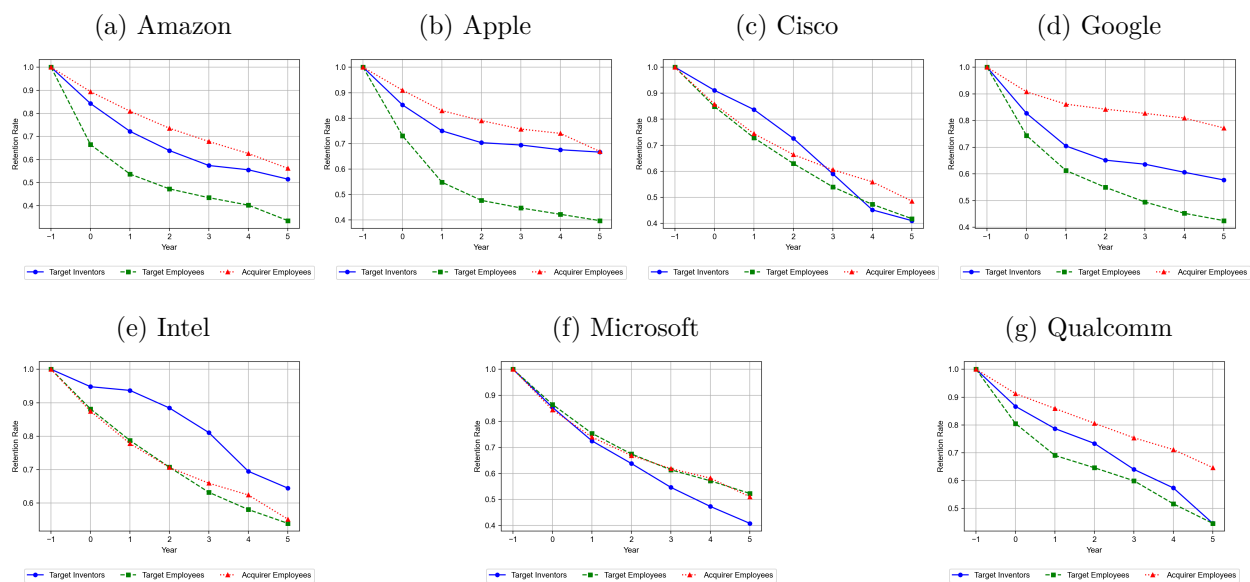


Table B.6: Innovation Heterogeneity by Number of Events

	(1)	(2)
	Unmatched	Matched sample
Post first Acq - No Follow-on	-0.04 (0.09)	-0.16 (0.11)
Post second Acq - No Follow-on	-0.31*** (0.12)	-0.36*** (0.13)
Post third Acq - No Follow-on	-0.31** (0.14)	-0.34** (0.15)
Post fourth Acq - No Follow-on	0.02 (0.17)	-0.04 (0.17)
Post first Acq - With Follow-on	0.43*** (0.05)	0.32*** (0.06)
Post second Acq - With Follow-on	0.20*** (0.05)	0.17*** (0.06)
Post third Acq - With Follow-on	0.15*** (0.06)	0.15** (0.07)
Post fourth Acq - With Follow-on	0.15* (0.08)	0.10 (0.11)
Constant	2.75*** (0.02)	4.53*** (0.05)
CPC FE	✓	✓
Year FE	✓	✓
Adj. R2	0.07	0.14
N	32968	16120

Standard errors clustered by CPC group in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.7: Grant Year Citations by subgroups of companies

	All cites (1)	From acquirer (2)	Inventor destinations (3)	Other employee destinations (4)	From neither (5)
PostAcquisition	0.303*** (0.042)	1.993*** (0.139)	0.265*** (0.081)	0.390*** (0.062)	0.138** (0.058)
Patent FE	✓	✓	✓	✓	✓
CPC×Year FE	✓	✓	✓	✓	✓
Pre-acq mean	0.576	0.022	0.066	0.169	0.478
Pseudo- R^2	0.579	0.578	0.578	0.578	0.578
Observations	35,573,144	35,523,748	35,516,751	35,548,506	35,560,466

Standard errors clustered by patent in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.8: Grant Year Citation effect by inventor retention rate at the acquirer

	All cites (1)	From acquirer (2)	Inventor destinations (3)	Other employee destinations (4)	From neither (5)
PostAcquisition	0.525*** (0.061)	2.567*** (0.185)	0.315*** (0.082)	0.500*** (0.077)	0.381*** (0.075)
PostAcquisition × Inventor Retention	-0.170 (0.146)	1.600*** (0.463)	-0.134 (0.188)	0.338** (0.168)	-0.497*** (0.183)
Patent FE	✓	✓	✓	✓	✓
CPC×Year FE	✓	✓	✓	✓	✓
Pre-acq mean	0.608	0.014	0.063	0.148	0.556
Pseudo- R^2	0.578	0.578	0.578	0.578	0.578
Observations	35,514,481	35,497,514	35,492,269	35,503,123	35,508,638

Standard errors clustered by patent in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.9: Mean patents per inventor-year, pre- vs post-acquisition, by status

Group	All patents		At acquirer		Inventors
	Pre	Post	Pre	Post	
Stayed at acquirer	0.893	1.379	0.129	1.130	738
Left after acquisition	0.993	0.937	0.098	0.523	288
Left before acquisition	0.703	0.649	0.049	0.167	469